TRAINER SLIDES · WSQ

# Effective Project Management for Small Projects

WSQ Course Code: TGS-2020505545 ·

2 days · 16 training hours

TSC: Project Management · ICT-PMT-3026-1.1

Version v19.0 · 19 August 2026

Tertiary Infotech Academy Pte Ltd · UEN: 201200696W



COURSE ADMINISTRATION

# Welcome & Housekeeping

How the two days work and what evidence you will produce

# Digital Attendance (Mandatory)

### AM, PM and Assessment

Complete every required SSG digital attendance scan.

### Scan the trainer QR

Use your mobile camera and submit the attendance record.

### 75% minimum

Attendance eligibility is calculated from the official record.

### Assessment required

Complete the assessment and achieve Competent for funding eligibility.

# Your Trainer

**Name**

Dr Alfred Ang

**Role**

Course developer and principal trainer, Tertiary Infotech Academy

**Experience**

20+ years across engineering, analytics, AI and digital transformation

**Contact**

enquiry@tertiaryinfotech.com · +65 6100 0613

# Ground Rules

### Participate

Use the tools, challenge assumptions and explain your decisions.

### Respect

One conversation at a time; disagree with evidence, not people.

### Be punctual

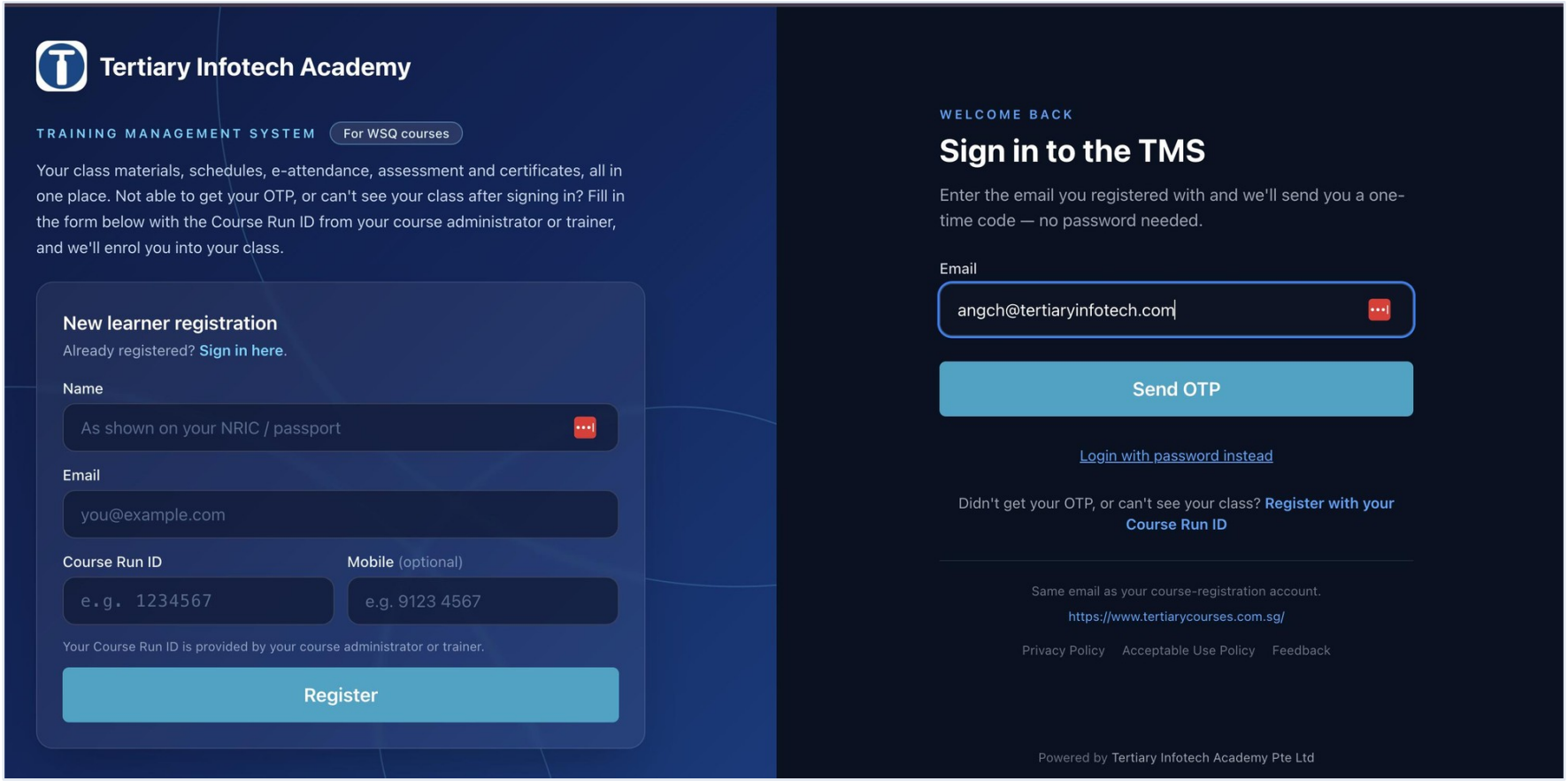
Activities build on one another; missed work creates downstream gaps.

### Keep evidence

Save every completed template for the case study and assessment review.

COURSE PORTAL

# Login to LMS and Download Course Material




Log in at lms-tms.tertiaryinfotech.com with your registered email and the OTP sent to your inbox, then open this course to download the slides and Learner Guide.

# Learning Outcomes — 1 to 4

**LO1**

Develop project scope and team to initiate the facilitation of a small project

**LO2**

Develop a realistic project plan based on the agreed project scope

**LO3**

Utilise appropriate methods and tools to track and drive a project

**LO4**

Identify, analyse and manage project risks

# Learning Outcomes — 5 to 7

**LO5**

Collaborate effectively with project stakeholders

**LO6**

Deploy manpower, budget and other resources for efficient completion

**LO7**

Track project deliverables and correct misalignments through control and closeout

# Skills Framework — Abilities A1 to A4

| Code | Ability | Practice |
|---|---|---|
| **A1** | Facilitate execution of small projects that are typically less than six months, with limited budget and interdependencies, and no significant strategic impact | Activities 1–2 |
| **A2** | Implement realistic project plans based on project objectives and scope | Activities 3–5 |
| **A3** | Utilise appropriate methods and tools to track and drive progress against plans and timelines | Activities 6 and 10 |
| **A4** | Identify risks to project success and take appropriate actions to manage them | Activity 7 |

**Every assessed ability has named tool-based evidence across the lifecycle.**

# Skills Framework — Abilities A5 to A7

| Code | Ability | Practice |
|---|---|---|
| **A5** | Collaborate effectively with relevant internal and external stakeholders directly impacting the project | Activity 8 |
| **A6** | Deploy manpower, financial budgets and relevant resources for efficient and effective completion | Activity 9 |
| **A7** | Track deliverables against schedules, monitor costs, timescales and resources, and take basic corrective actions when misaligned | Activities 11–12 |

**The twelve activities form one connected project evidence trail.**

# Skills Framework — Knowledge

| Code | Knowledge | Taught in |
|---|---|---|
| **K1** | Elements of a small project | Topic 1 |
| **K2** | Requirements of a project plan | Topic 2 |
| **K3** | Application of appropriate project management methodologies and tools | Topic 3 |
| **K4** | Project risks | Topic 4 |
| **K5** | Project stakeholder identification | Topic 5 |

**Knowledge is applied in the activity immediately following its topic.**

# Lesson Plan — Day 1

### Initiate with value

Topic 1 · charter, boundaries, team and kickoff

### Plan the work

Topic 2 · requirements, WBS, schedule, critical path and budget

### Drive quality and flow

Topic 3 · visual management, root-cause and prioritisation tools

### Manage uncertainty

Topic 4 · risk register, responses, contingency and decision analysis

# Lesson Plan — Day 2

### Engage stakeholders

Topic 5 · mapping, RACI, communication and engagement

### Deploy resources

Topic 6 · capacity, optimisation, procurement and team performance

### Control and close

Topic 7 · status, earned value, change, corrective action and handover

### Assessment

Written Case Study and Oral Questioning, open book

# Course Outline — Topics 1 to 4

### Topic 01 · Initiation of a Project

Small-project fit · value · charter · scope · team · kickoff · A1, K1

### Topic 02 · Develop a Project Plan

Requirements · scope baseline · WBS · network · schedule · budget · A2, K2

### Topic 03 · Apply Methods and Tools to Drive a Project

Quality planning · flow · root cause · Pareto · improvement · A3, K3

### Topic 04 · Manage Project Risk

Identify · assess · respond · contingency · monitor · A4, K4

# Course Outline — Topics 5 to 7

### Topic 05 · Collaborate with Stakeholders

Identify · analyse · communicate · engage · allocate accountability · A5, K5

### Topic 06 · Deploy Resources

People · budget · equipment · procurement · team performance · A6

### Topic 07 · Track Project Deliverables and Correct Misalignments

Performance · earned value · change · corrective action · handover · closure · A7

# The Twelve Tool Packs

**Activity 1**
Selection & Cost-Benefit — Screen and select the project

**Activity 2**
Business Case & Charter — Authorise and kick off

**Activity 3**
Requirements Workshop & RTM — Discover and trace needs

**Activity 4**
WBS & Dictionary — Build the scope baseline

**Activity 5**
Network, Gantt & Budget — Integrate schedule and cost

**Activity 6**
Delivery & Quality Plan — Tailor execution and quality

**Activity 7**
Risk Decision Tools — Fund risk responses

**Activity 8**
Stakeholder, RACI & SEAM — Align decisions and engagement

**Activity 9**
Capacity & Procurement — Deploy resources

**Activity 10**
Root-Cause Toolchain — Diagnose quality

**Activity 11**
Status, EVM, Issue & Change — Control performance

**Activity 12**
Acceptance, Handover & Benefits — Close and transfer

# One Small Project Runs Through All Twelve Activities

### Scope

A three-outlet click-and-collect pilot for HarbourHub Retail.

### Time & money

Five months, a S$180,000 cap and a seven-person core team.

### Interfaces

Two vendors plus outlet operations, finance, IT and customers.

### Success

Launch on date; 120 orders/week; 98% accuracy; 4.2/5 satisfaction; within cap.

# Briefing for Assessment

### Individual work

No discussion once assessment begins.

### Open book

Slides, Learner Guide and approved materials only.

### Protect the paper

No photographs, recording or unauthorised materials.

### Answer with evidence

State the method, decision, rationale and expected result.

# Final Assessment

### Written Case Study

Written Assessment (Case Study) — apply project management judgement to a small-project scenario.

### Oral Questioning

Oral Questioning — explain and justify decisions against the course methods and evidence.

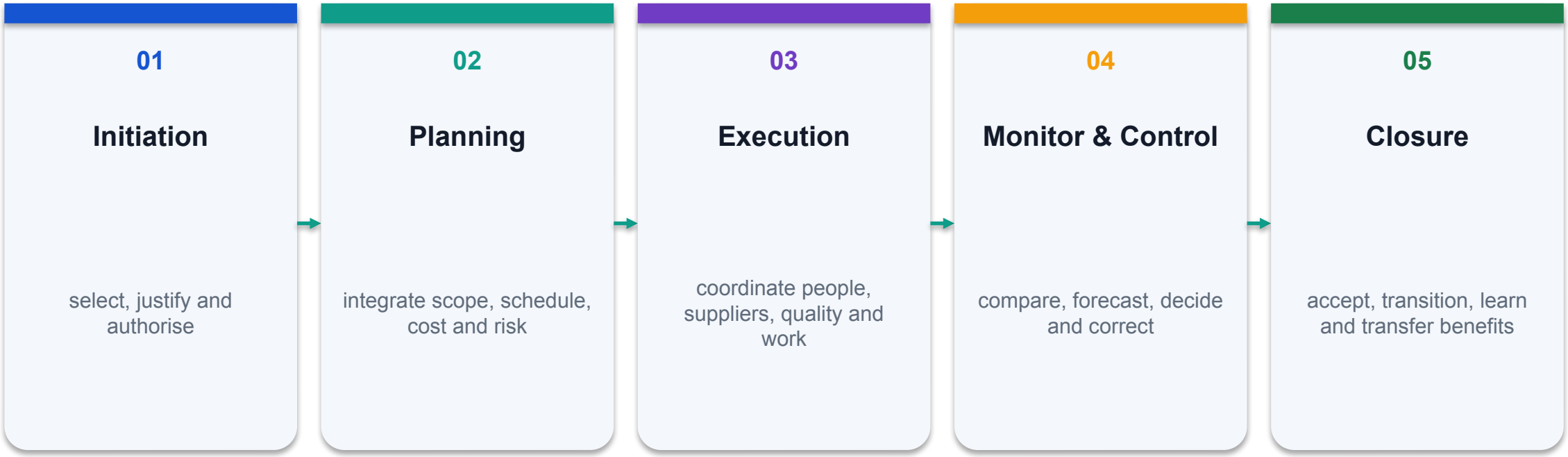
### Conditions

Open book: course slides, Learner Guide and approved course materials only.

### Competency

A minimum of 75% attendance and a Competent result are required for funding eligibility.

# The Complete Project Lifecycle




**The phases overlap in practice; control and engagement operate from authorisation to closure.**

# Lifecycle Phases and WSQ Topics

| Phase | Primary course topics | Evidence produced |
|---|---|---|
| **Initiation** | Topic 1 | Selection, business case, charter, kickoff |
| **Planning** | Topics 2 and 4 | Requirements, baselines, risk responses |
| **Execution** | Topics 3, 5 and 6 | Work, quality, engagement, resources |
| **Monitoring & Controlling** | Topics 3, 4, 5 and 7 | Status, forecasts, issues, change, action |
| **Closure** | Topic 7 | Acceptance, handover, lessons, benefits ownership |

**The seven-topic syllabus is preserved while the lifecycle makes the end-to-end operating system visible.**

# What Runs Across Every Phase

### Integration

Keep decisions and artefacts consistent.

### Stakeholders

Engage before decisions become resistance.

### Risk

Refresh uncertainty as facts change.

### Quality

Define, assure and control fitness for use.

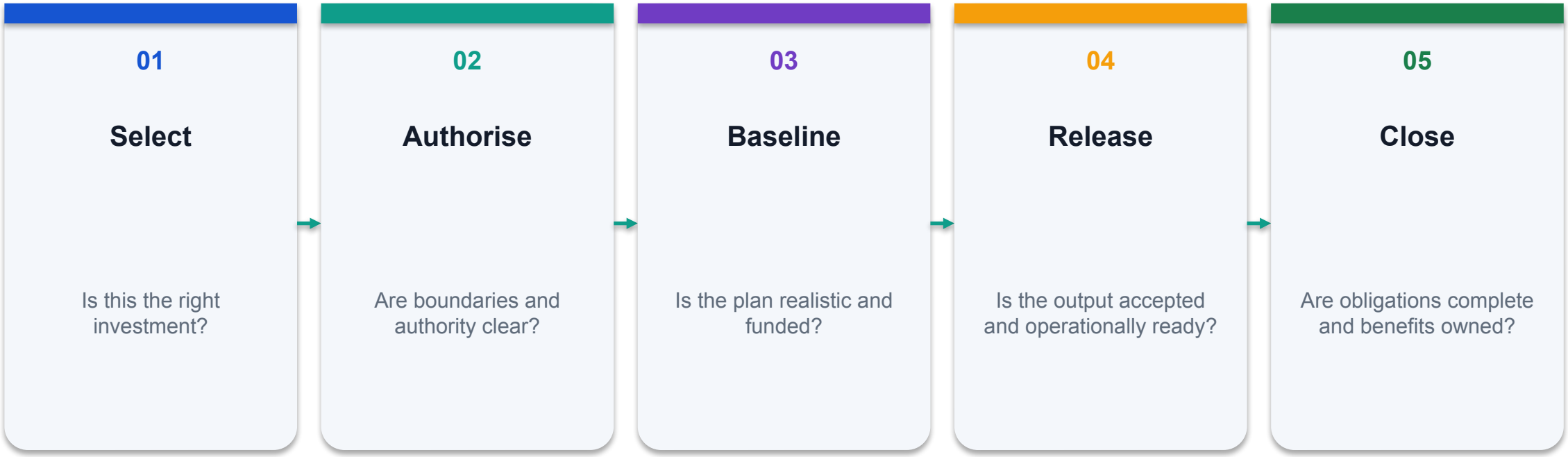
### Resources

Match skills and capacity to priority work.

### Governance

Escalate outside agreed tolerances.

# Five Phase Gates for a Small Project



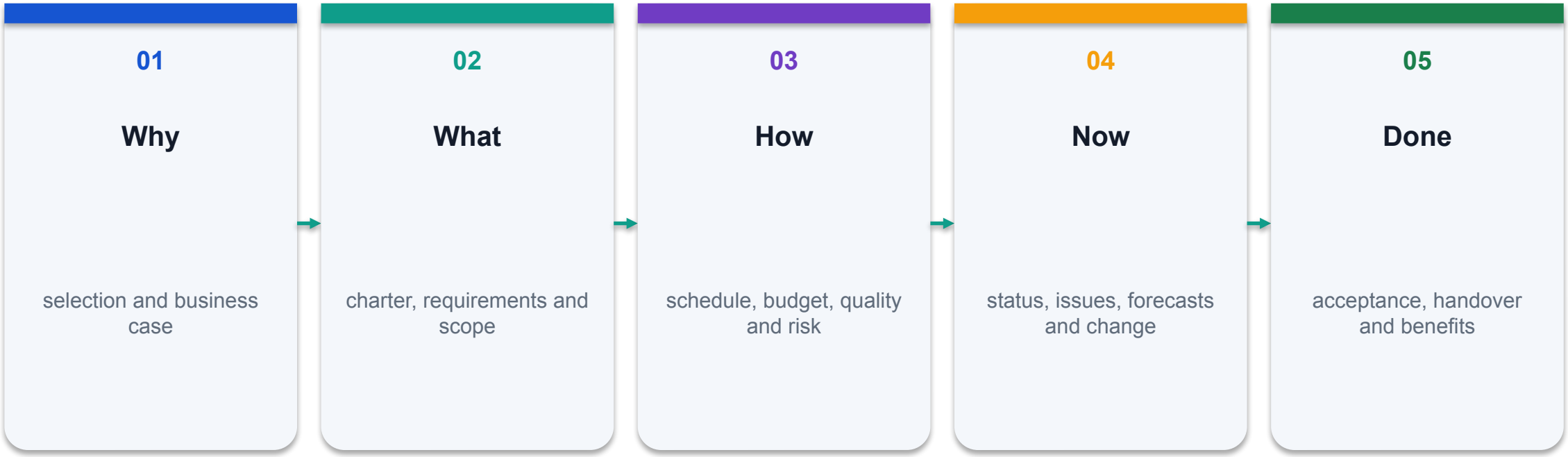

**A gate is a decision with evidence and conditions—not a ceremonial meeting.**

# Tailoring Control to a Small Project

| Need | Lean control | Escalate when |
|---|---|---|
| **Scope** | One charter, WBS and RTM | Success criteria or exclusions move |
| **Schedule** | Milestones plus critical path | Tolerance or launch date is threatened |
| **Cost** | Time-phased budget | Forecast exceeds authority |
| **Risk** | Prioritised register | Trigger, exposure or reserve changes materially |
| **Change** | One log and named approver | Trade-off crosses a tolerance |

**Tailoring removes low-value ceremony while preserving decision integrity.**

# The Connected Evidence Chain




**Each artefact must feed the next decision; isolated templates are administration, not management.**

# Twelve Activities, One Project

**Initiate**

Activities 1–2

**Plan**

Activities 3–5 and 7

**Execute**

Activities 6, 8 and 9

**Control**

Activities 10–11

**Close**

Activity 12

01

TOPIC 01 · A1, K1 · LO1

# Initiation of a Project

Small-project fit · value · charter · scope · team · kickoff

# Small does not mean informal.

**The smaller the project, the less waste it can absorb. A short charter, sharp boundaries and visible decision rights are the control system—not bureaucracy.**

# Initiation of a Project — Core Concepts I

**01**

A project is temporary, unique and value-seeking; operations are ongoing and repetitive.

**02**

A small project is normally under six months, has a limited budget and team, few dependencies and no significant strategic impact.

**03**

The project manager integrates people, decisions and trade-offs; they do not merely maintain a schedule.

**04**

Success balances value and outcomes with scope, schedule, cost, quality, resources and risk.

# Initiation of a Project — Core Concepts II

**05**

Predictive, adaptive and hybrid approaches are choices driven by uncertainty, not labels of maturity.

**06**

The business case tests whether the project should exist; the charter authorises it to exist.

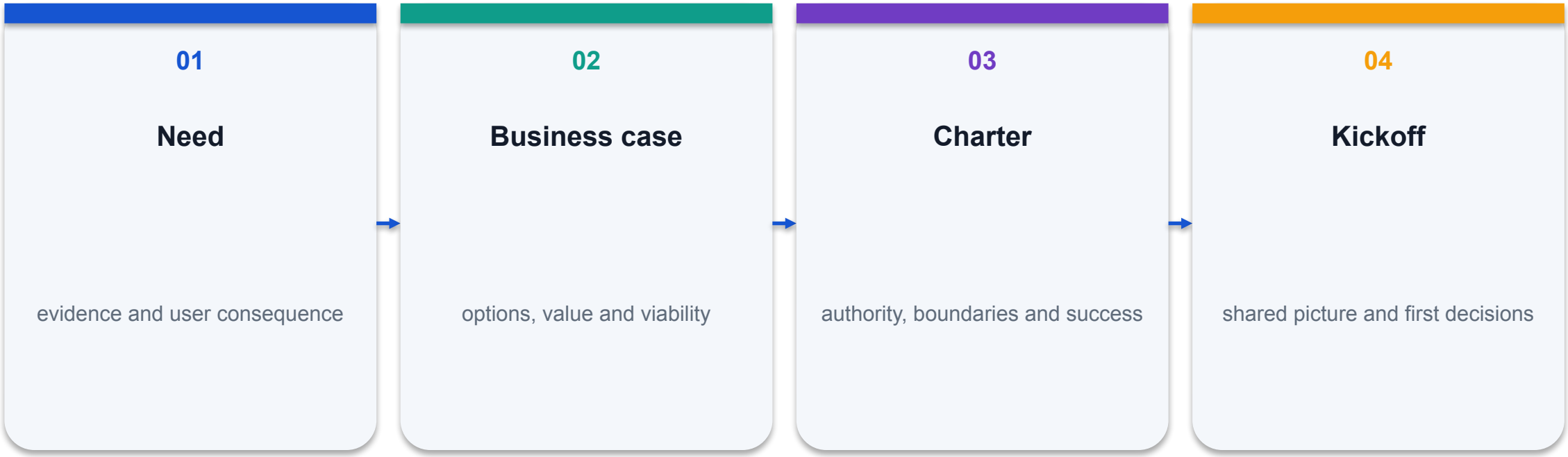
**07**

A useful charter names purpose, measurable success, scope boundaries, milestones, budget, major risks, team and sponsor authority.

**08**

A kickoff creates one shared picture of why, what, who, how and what happens next.

# From Need to Authorised Work




**Do not plan an unauthorised idea as though it were already a project.**

# Business Case vs Project Charter

| Dimension | Business Case | Project Charter |
|---|---|---|
| **Core question** | Should we invest? | Are we authorised to proceed? |
| **Focus** | Options, benefits, costs, risk | Purpose, scope, team, milestones |
| **Owner** | Sponsor / business owner | Sponsor with project manager |
| **Decision** | Select or reject option | Authorize and delegate |

**Both are living decision records; neither is a ceremonial form.**

# Choosing a Development Approach

| Signal | Predictive | Adaptive / Hybrid |
|---|---|---|
| **Requirements** | Stable and testable upfront | Evolving or discovery-heavy |
| **Feedback** | Available late or at gates | Available in short cycles |
| **Change cost** | High after commitment | Reduced through increments |
| **Governance** | Baseline and stage gates | Timeboxes plus control points |

**Approach follows uncertainty and reversibility—not fashion.**

# Project Work versus Operations

| Dimension | Project | Operations |
|---|---|---|
| **Purpose** | Create a unique result | Sustain recurring value |
| **Duration** | Temporary | Ongoing |
| **Change** | Introduces change | Stabilises and improves work |
| **Closure** | Formal endpoint | Continues until redesigned |

**The same people may do both; the management system changes with the nature of the work.**

# The Small-Project Qualification Test

### Time

Normally less than six months.

### Budget

Limited enough for proportionate governance.

### Team

Small core team with clear interfaces.

### Dependencies

Few and manageable.

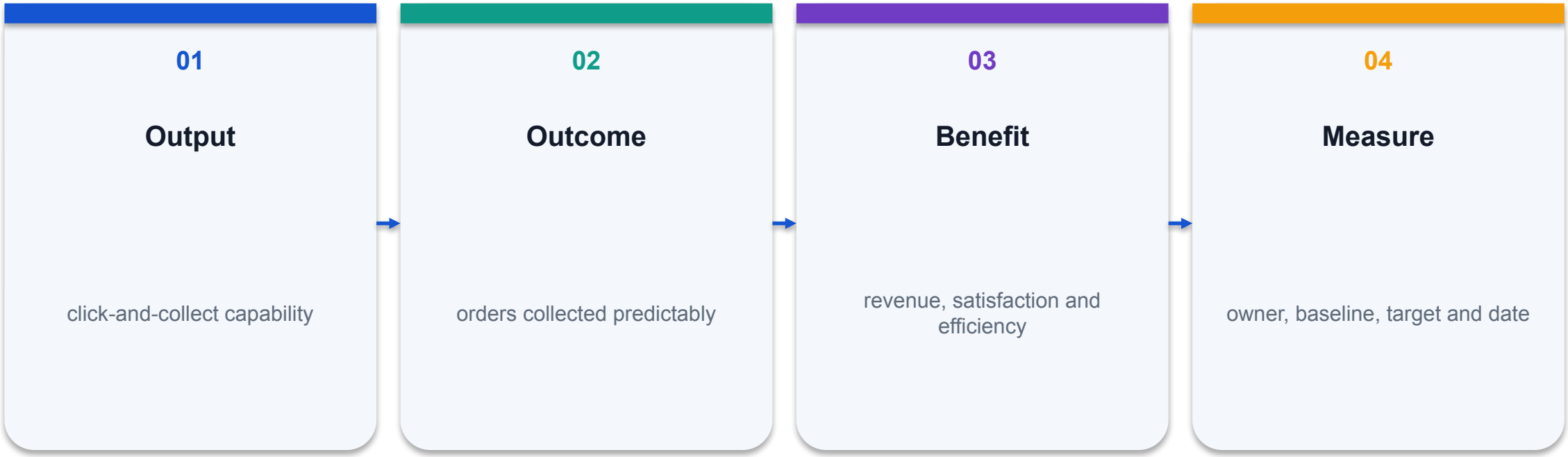
### Impact

No significant strategic or enterprise-wide effect.

### Uncertainty

Can be bounded through short learning cycles.

# From Output to Benefits




**Deliverables are produced by the project; benefits are usually realised with operations.**

# Project Selection Methods

| Method | Best for | Watch for |
| --- | --- | --- |
| **Cost-benefit** | Financial viability | Unsupported benefit assumptions |
| **Weighted scoring** | Multiple criteria | Biased weights |
| **Payback** | Liquidity and speed | Ignores later value |
| **Mandatory driver** | Compliance or safety | Treating obligation as optional |

**Use more than one lens when a small financial difference hides a major operational consequence.**

# Sponsor and Project Manager—Different Accountability

### Sponsor

Owns investment, priority and escalation.

### Project manager

Integrates delivery and advises decisions.

### Business owner

Owns adoption and benefits after handover.

### Functional manager

Provides capability and resolves resource conflicts.

# Business Documents at Initiation

| Artefact | Decision supported | Owner |
|---|---|---|
| **Business case** | Invest, hold or reject | Sponsor / business owner |
| **Benefits plan** | How value will be measured | Benefits owner |
| **Project charter** | Authorise and delegate | Sponsor with PM |
| **Assumptions log** | What must be validated | Project manager |

**Keep documents brief, but never combine ownership so completely that accountability disappears.**

# A Charter That Can Govern

### Purpose

The problem and desired outcome.

### Success

Measures, targets and dates.

### Boundaries

Deliverables and exclusions.

### Authority

Sponsor, PM and tolerances.

### Frame

Milestones, budget and constraints.

### Exposure

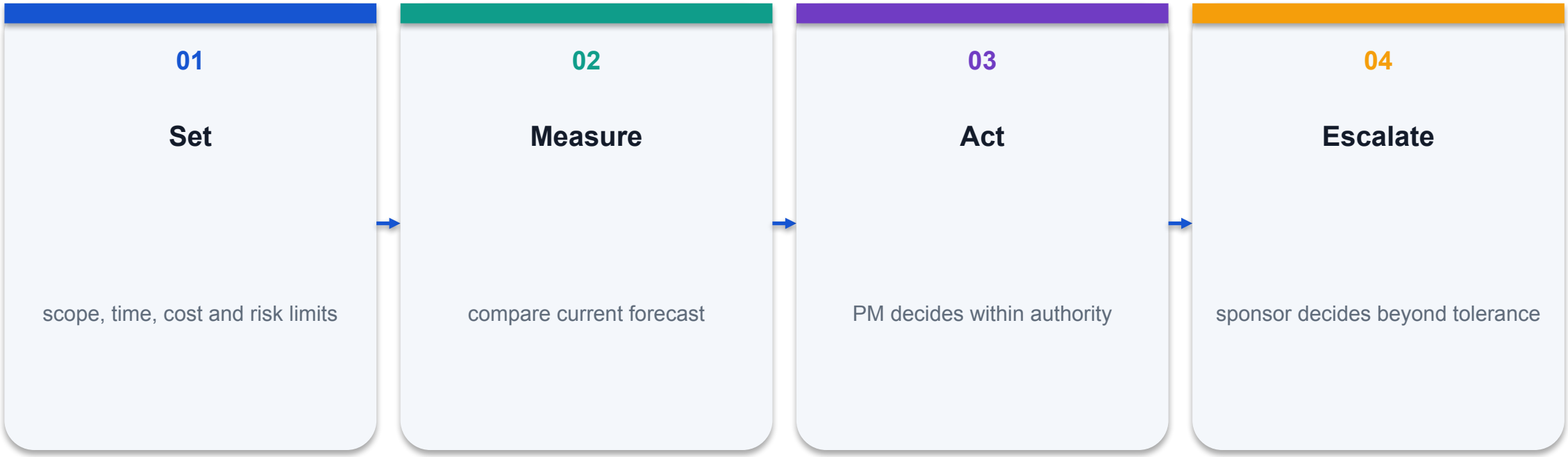
Assumptions, dependencies and high risks.

# Assumption, Constraint, Dependency or Risk?

| Type | Meaning | Example |
|---|---|---|
| **Assumption** | Believed true for planning | Outlet Wi-Fi capacity is sufficient |
| **Constraint** | Fixed limitation | S$180,000 ceiling |
| **Dependency** | Reliance on other work | POS API access before integration |
| **Risk** | Uncertain event and impact | API access may be late |

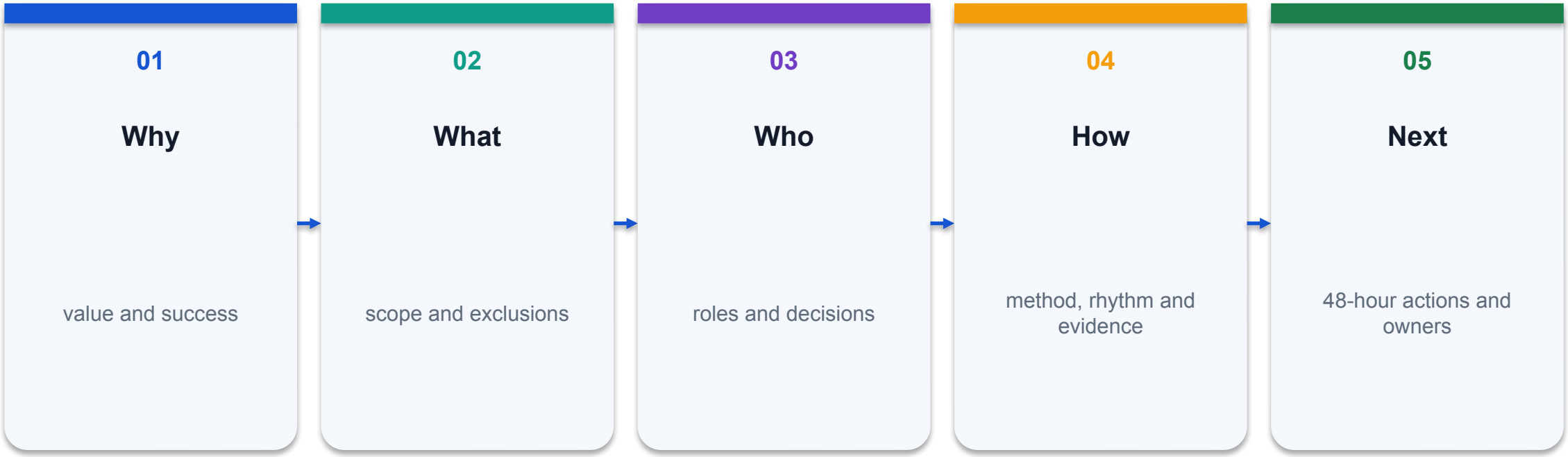
**Classify correctly because each item needs a different response.**

# Lightweight Governance with Tolerances




**Authority without tolerances creates hesitation; tolerances without reporting create surprise.**

# Kickoff as an Alignment Decision




**A good kickoff reveals mismatched assumptions early enough to repair them cheaply.**

# Activity 1 Tool Pack

XLSX

**02 Cost Benefit Analysis**

Editable evidence file

XLSX

**01 Small Project Screening and Selection**

Editable evidence file

DECISION ENABLED

**A completed small-project screen, weighted option scorecard, cost-benefit check and sponsor-ready selection recommendation.**

ACTIVITY 1 · 35 MIN

# Screen and Select the HarbourHub Pilot

## THE SITUATION

HarbourHub Fresh has three competing improvement ideas: click-and-collect, a loyalty app, and a warehouse automation study. Only one can be funded this quarter. The sponsor needs a transparent selection decision and evidence that the selected work fits the small-project profile.

## WHAT YOU'LL DO

Apply the small-project screen, compare options using weighted criteria, test the financial case and document the recommendation, assumptions and rejection reasons.

## YOU'LL PRODUCE

A completed small-project screen, weighted option scorecard, cost-benefit check and sponsor-ready selection recommendation.

## DONE WHEN

The selected option satisfies every small-project criterion, the weighted score reconciles, the financial downside is visible and the recommendation states a clear gate decision.

**Tool: Small Project Screening, Multi-Criteria Selection and Cost-Benefit Analysis**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 1 folder.

# Debrief — Activity 1: What It Proves

1

Selection prevents the team from efficiently delivering the wrong project.

2

Weights expose priorities; narrative rationale prevents false precision.

3

A selected idea is still not authorised work until the charter gate is passed.

# Activity 2 Tool Pack

DOCX

**05 Business Case on a Page**

Editable evidence file

XLSX

**08 Project Charter Template**

Editable evidence file

DECISION ENABLED

**A one-page business case, signed-ready charter, assumptions and constraints log, milestone frame, decision tolerances and kickoff action list.**

ACTIVITY 2 · 45 MIN

# Authorise the Pilot and Run the Kickoff Gate

## THE SITUATION

The click-and-collect pilot has been selected, but Operations, Finance and the two vendors hold different assumptions about success, exclusions and decision authority. The sponsor will authorise delivery only after a concise gate review.

## WHAT YOU'LL DO

Convert the selected option into a measurable business case and charter, define boundaries, name decision rights and conduct a short sponsor gate and kickoff.

## YOU'LL PRODUCE

A one-page business case, signed-ready charter, assumptions and constraints log, milestone frame, decision tolerances and kickoff action list.

## DONE WHEN

The charter names one accountable sponsor, measurable success, explicit exclusions, decision tolerances and an unambiguous authorisation decision within the small-project constraints.

**Tool: Business Case on a Page and Project Charter**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 2 folder.

# Debrief — Activity 2: What It Proves

1

The business case justifies investment; the charter grants authority.

2

Decision tolerances make lightweight governance possible.

3

Kickoff converts signed paper into shared commitments and immediate action.

# Topic 01 Recap

### Decision

A project is temporary, unique and value-seeking; operations are ongoing and repetitive.

### Method

Success balances value and outcomes with scope, schedule, cost, quality, resources and risk.

### Evidence

A one-page business case, signed-ready charter, assumptions and constraints log, milestone frame, decision tolerances and kickoff action list.

### Capability

Define scope, success, authority and team arrangements for an executable small-project start (A1, K1).

TOPIC 02 · A2, K2 · LO2

# Develop a Project Plan

Requirements · scope baseline · WBS · network · schedule · budget

# A plan is a connected model, not a collection of files.

**Requirements, WBS, dependencies, dates, resources and money must describe the same work. If one changes, the others must move with it.**

# Develop a Project Plan — Core Concepts I

01

The plan integrates subsidiary decisions; separate scope, schedule and cost files are not yet a project plan.

02

Progressive elaboration and rolling-wave planning add detail as uncertainty reduces without abandoning control.

03

Requirements must be testable, prioritised and traceable from need through deliverable and acceptance evidence.

04

The scope statement defines deliverables, exclusions, constraints, assumptions and acceptance criteria.

# Develop a Project Plan — Core Concepts II

**05**

The WBS decomposes deliverables to manageable work packages; the WBS dictionary removes ambiguity.

**06**

Dependencies create the network; the longest path sets the earliest finish and carries zero or least float.

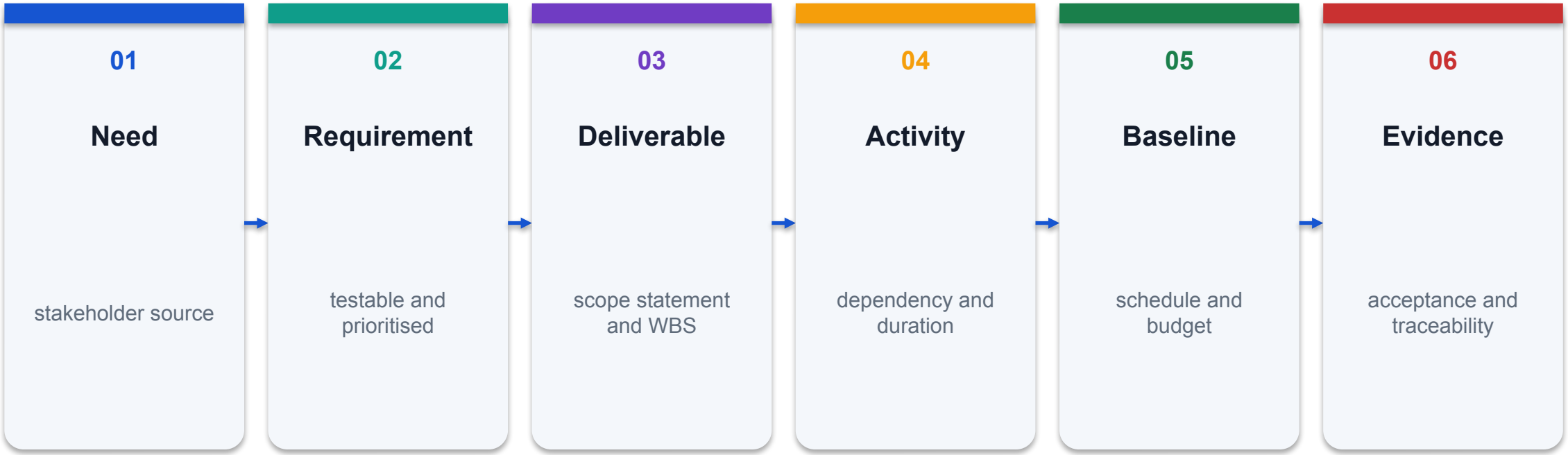
**07**

Duration estimates should separate work, waiting, risk and resource availability; use analogous, parametric, bottom-up or three-point methods deliberately.

**08**

The cost baseline is time-phased; contingency reserve covers known-unknowns while management reserve covers unknown-unknowns.

# The Planning Chain



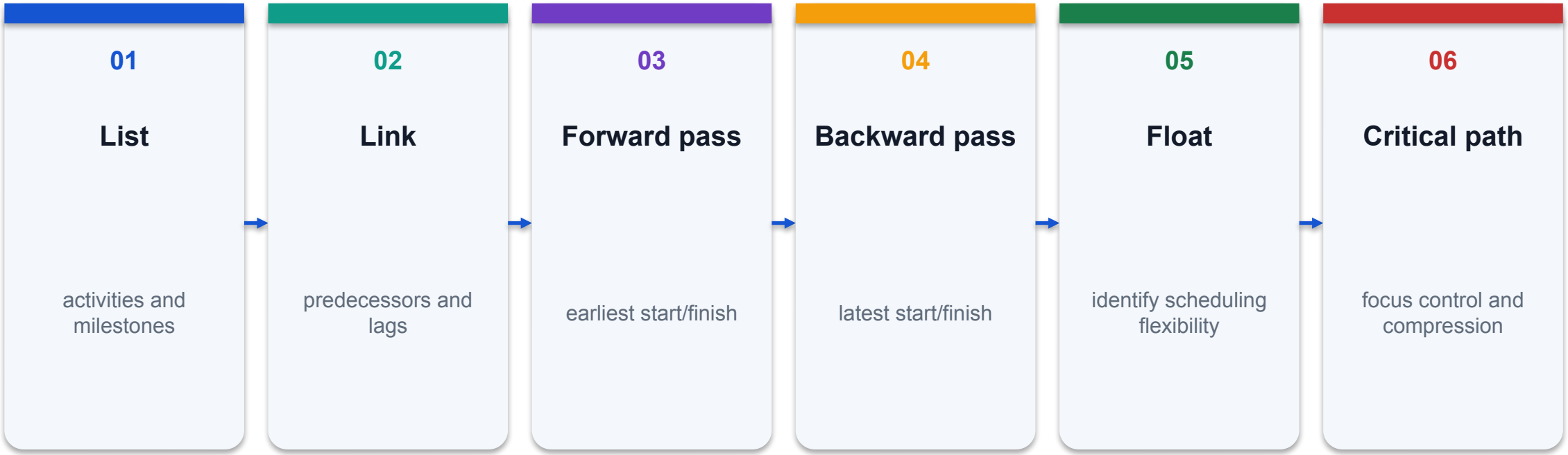

**A broken link creates either orphan scope or unplanned work.**

# Estimating Methods

| Method | Best use | Main risk |
|---|---|---|
| **Analogous** | Early estimate from similar work | False similarity |
| **Parametric** | Repeatable unit rates | Bad rate or scale |
| **Bottom-up** | Defined work packages | Time and detail cost |
| **Three-point** | Uncertain durations/costs | Optimism in inputs |

**Record the basis of estimate so future variance can be explained.**

# Critical Path Logic




**Recalculate after every approved change—the critical path can move.**

# The Integrated Project Plan

### Scope

What is authorised and accepted.

### Schedule

Sequence, duration and milestones.

### Cost

Time-phased funding and reserves.

### Quality

Measures, assurance and control.

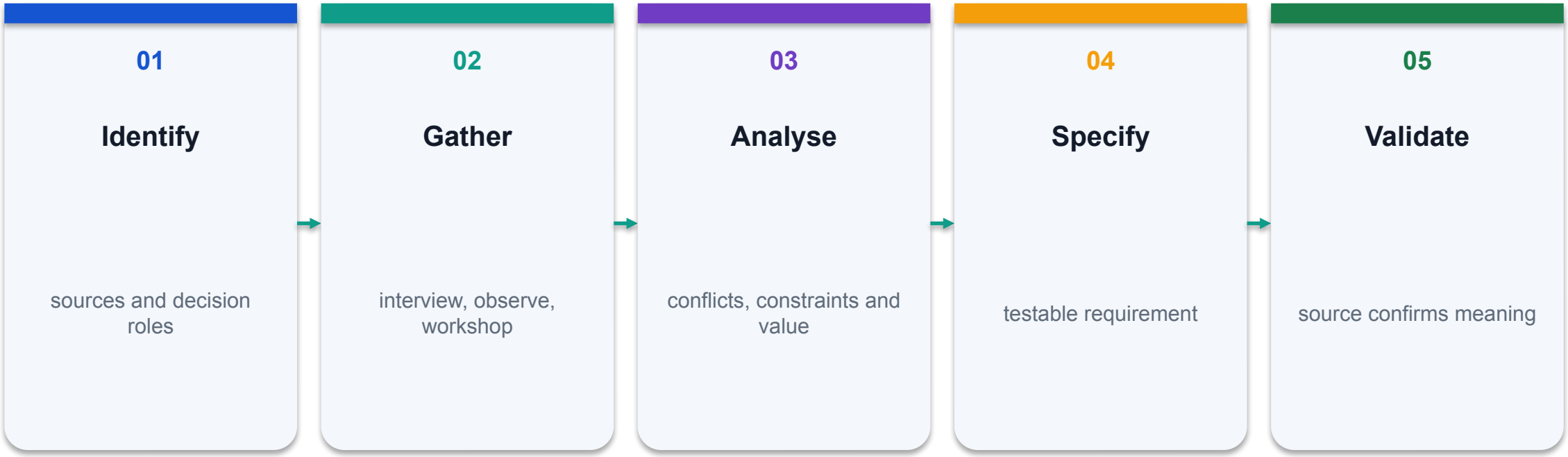
### Resources

Skills, capacity and procurement.

### Risk & stakeholders

Responses, engagement and governance.

# Requirements Elicitation




**Elicitation is discovery and negotiation—not taking an order from one stakeholder.**

# Weak versus Testable Requirements

| Weak wording | Testable wording | Evidence |
| --- | --- | --- |
| **Fast collection** | 90% ready within 10 minutes | Timestamp sample |
| **Accurate picking** | At least 98% line accuracy | Check sheet |
| **Easy to use** | New staff completes task after 30-minute training | Observed test |
| **Reliable** | 99.5% availability in pilot hours | System log |

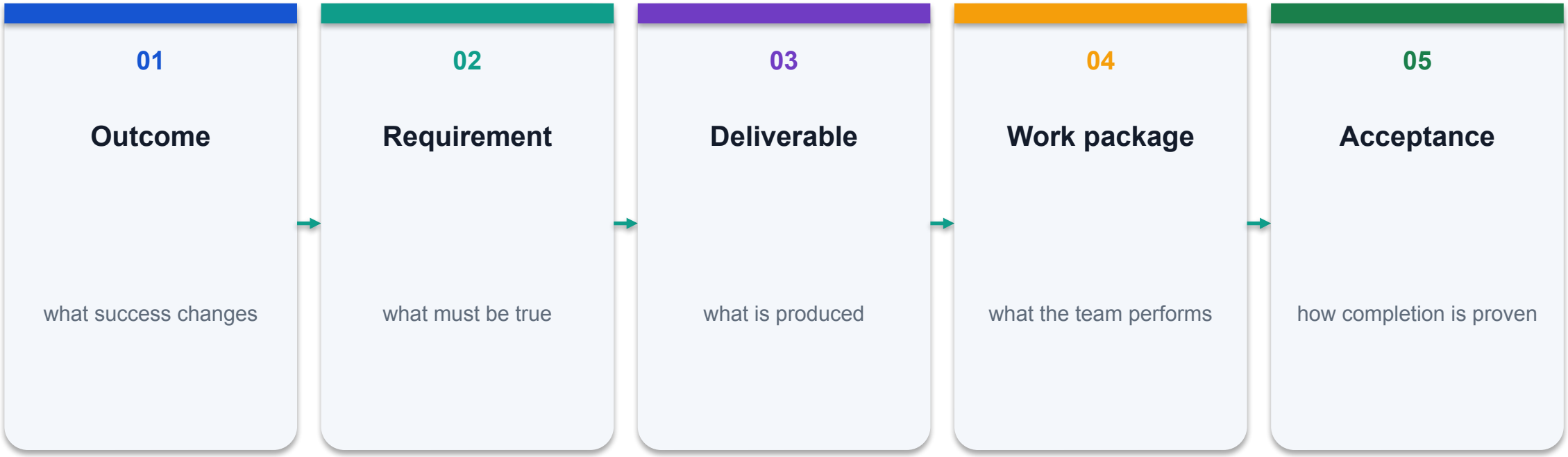
**A measure without a method or status date is still ambiguous.**

# Prioritisation Choices

| Method | Decision logic | Use when |
|---|---|---|
| **MoSCoW** | Must / Should / Could / Won't | Time-boxed scope |
| **Weighted scoring** | Value across criteria | Competing options |
| **Kano** | Basic, performance, delight | Customer experience |
| **Trade-off sliders** | Relative constraint priority | Governance alignment |

**Priorities must remain traceable to outcomes, not just stakeholder preference.**

# Traceability from Need to Evidence




**Trace both directions: every need is delivered, and every deliverable has a reason.**

# Scope Statement Essentials

### Product scope

Characteristics of the result.

### Project scope

Work required to create it.

### Exclusions

What is deliberately not included.

### Acceptance

Who approves against which criteria.

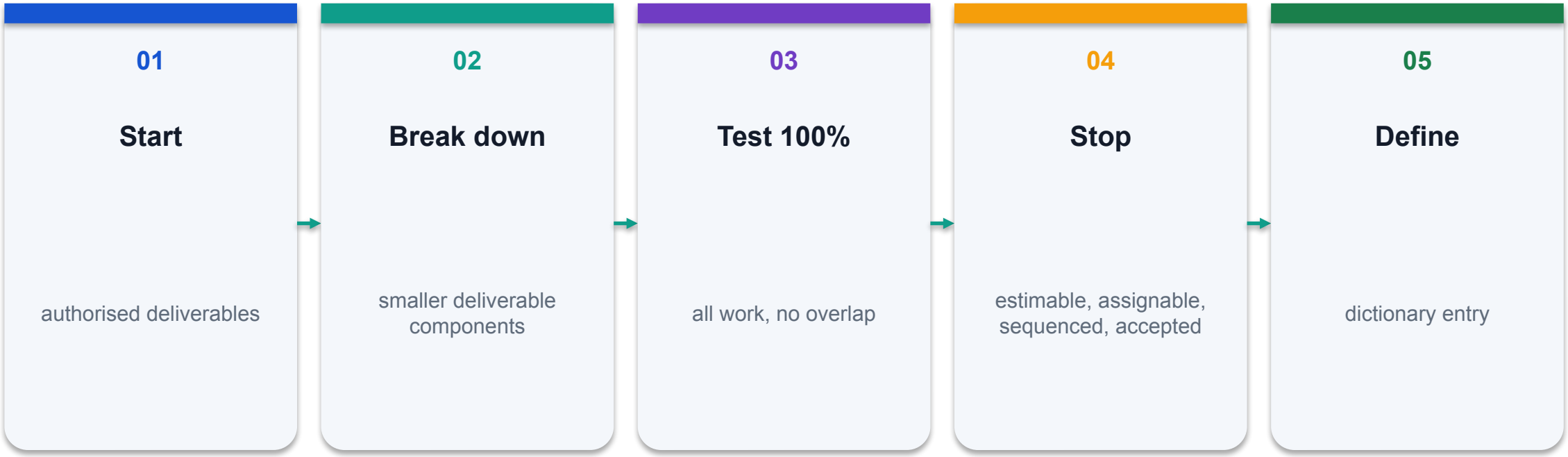
### Constraints

Limits the plan must respect.

### Assumptions

Planning beliefs to validate.

# WBS Decomposition




**Do not decompose by department when the deliverable crosses departments.**

# WBS Dictionary—Why It Matters

| Field | Decision enabled | Failure prevented |
|---|---|---|
| **Description** | Shared meaning | Different estimates for different work |
| **Owner** | Accountability | Unclaimed delivery |
| **Acceptance** | Definition of complete | Almost done forever |
| **Dependencies** | Sequence | Late discovery |
| **Estimate basis** | Confidence and revision | Unexplained padding |

**The diagram shows structure; the dictionary makes the structure manageable.**

# Dependency Relationships

| Type | Logic | Example |
|---|---|---|
| **Finish-to-start** | A finishes before B starts | Configure before integration test |
| **Start-to-start** | A starts before B starts | Draft training after testing begins |
| **Finish-to-finish** | A finishes before B finishes | Documentation completes with testing |
| **Start-to-finish** | Rare transition logic | Old support ends after new starts |

**Leads and lags change timing; they do not remove the underlying dependency.**

# Critical Path Calculation

**01**

**Forward pass**

earliest start and finish

**02**

**Project finish**

largest earliest finish

**03**

**Backward pass**

latest finish and start

**04**

**Float**

late start minus early start

**05**

**Critical path**

least-float path

**The critical path is the longest-duration route through current logic, not the most important-looking task.**

# Estimating Methods

| Method | Strength | Limitation |
| --- | --- | --- |
| **Analogous** | Fast early estimate | Lower detail |
| **Parametric** | Repeatable unit logic | Needs valid relationship |
| **Bottom-up** | Detailed and accountable | Time-consuming |
| **Three-point** | Makes uncertainty visible | Still depends on judgement |

**Record the basis so the estimate can be challenged and updated without starting over.**

# Cost Baseline and Reserves

### Work-package estimates

Rolled up from authorised scope.

### Contingency reserve

Funded responses for identified risk.

### Cost baseline

Time-phased performance reference.

### Management reserve

Held outside baseline for unknown-unknowns.

### Project budget

Cost baseline plus management reserve.

# Activity 3 Tool Pack

XLSX

**25 Requirements Traceability Matrix**

Editable evidence file

XLSX

**03 Requirements Workshop and Stakeholder Register**

Editable evidence file

**DECISION ENABLED**

**A stakeholder register, workshop decision log and requirements traceability matrix with priorities, owners, deliverables and measurable acceptance evidence.**

ACTIVITY 3 · 35 MIN

# Discover Requirements and Map Acceptance Evidence

## THE SITUATION

Outlet supervisors want speed, Finance wants payment control, customers want predictable collection, and the POS vendor wants a stable interface. The charter is approved, but the stated needs are not testable.

## WHAT YOU'LL DO

Identify sources, facilitate a structured workshop, convert needs into testable requirements, prioritise with MoSCoW and define acceptance evidence and ownership.

## YOU'LL PRODUCE

A stakeholder register, workshop decision log and requirements traceability matrix with priorities, owners, deliverables and measurable acceptance evidence.

## DONE WHEN

Every prioritised requirement has a source, rationale, owner, deliverable and testable acceptance method; unresolved conflicts have named decision owners and dates.

**Tool: Requirements Workshop, Stakeholder Register and Traceability Matrix**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 3 folder.

# Debrief — Activity 3: What It Proves

1

A requirement is not complete until acceptance can be observed or measured.

2

Prioritisation is a trade-off decision, not a label applied by the loudest stakeholder.

3

Traceability protects value when scope changes later.

# Activity 4 Tool Pack

PPTX

**24 Work Breakdown Structure**

Editable evidence file

XLSX

**24 WBS Dictionary**

Editable evidence file

XLSX

**25 Requirements Traceability Matrix**

Editable evidence file

DECISION ENABLED

**A scope statement, WBS, WBS dictionary and repaired traceability matrix with no missing or duplicated work.**

ACTIVITY 4 · 45 MIN

# Build the Scope Baseline with WBS and Dictionary

## THE SITUATION

The team has 14 prioritised requirements, but the vendor proposal groups work by department and omits outlet readiness, training and handover. Estimates cannot be trusted until the total scope is visible.

## WHAT YOU'LL DO

Create a deliverable-oriented WBS, apply the 100% rule, define work packages in the dictionary and reconcile every requirement and acceptance criterion to scope.

## YOU'LL PRODUCE

A scope statement, WBS, WBS dictionary and repaired traceability matrix with no missing or duplicated work.

## DONE WHEN

The WBS covers 100% of authorised work without overlap, each work package is manageable and every requirement traces to acceptance-ready scope.

**Tool: Work Breakdown Structure, WBS Dictionary and Requirements Traceability Matrix**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 4 folder.

# Debrief — Activity 4: What It Proves

1

The WBS describes all project work; the schedule describes when that work occurs.

2

Transition and project-management work are real scope even when customers never see them.

3

A strong dictionary prevents different teams from estimating different meanings.

# Activity 5 Tool Pack

XLSX

**28 Schedule Network Diagram**

Editable evidence file

XLSX

**29 - Gantt Chart - Schedule - Daily Weekly Monthly and Phases**

Editable evidence file

XLSX

**32 Project Budget - Planned vs Actual - Weekly and Monthly**

Editable evidence file

**DECISION ENABLED**

**A network with forward/backward passes, critical path, Gantt milestones, basis of estimates and a time-phased cost baseline plus contingency within the authorised ceiling.**

ACTIVITY 5 · 55 MIN

# Develop the Integrated Schedule and Cost Baseline

## THE SITUATION

The launch must precede Lunar New Year trading. The vendor has proposed a date without dependency logic, resource calendars or reserves. Finance requires a time-phased baseline within S$180,000.

## WHAT YOU'LL DO

Sequence work packages, calculate the critical path and float, estimate duration and cost using explicit methods, set milestones and reconcile schedule, budget and reserve.

## YOU'LL PRODUCE

A network with forward/backward passes, critical path, Gantt milestones, basis of estimates and a time-phased cost baseline plus contingency within the authorised ceiling.

## DONE WHEN

The dependency network calculates correctly, the plan fits the five-month limit, all authorised work is time-phased and funded, and baseline plus contingency stays within S$180,000.

**Tool: Network Diagram, Gantt Schedule and Budget versus Actual**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 5 folder.

# Debrief — Activity 5: What It Proves

1

A Gantt is the presentation of schedule logic, not the logic itself.

2

The critical path can change after every update or approved change.

3

Baseline and forecast must remain separate so variance can be explained.

# Topic 02 Recap

### Decision

The plan integrates subsidiary decisions; separate scope, schedule and cost files are not yet a project plan.

### Method

The scope statement defines deliverables, exclusions, constraints, assumptions and acceptance criteria.

### Evidence

A network with forward/backward passes, critical path, Gantt milestones, basis of estimates and a time-phased cost baseline plus contingency within the authorised ceiling.

### Capability

Sequence, estimate and fund the authorised scope as one realistic plan (A2, K2).

TOPIC 03 · A3, K3 · LO3

# Apply Methods and Tools to Drive a Project

Quality planning · flow · root cause · Pareto · improvement

# Use the lightest tool that changes the next decision.

**A tool earns its place when it turns uncertain work into visible evidence: flow, cause, variation, priority or ownership.**

# Apply Methods and Tools to Drive a Project — Core Concepts I

**01**

A method defines how work is managed; a tool makes information visible. Choose both for the decision needed.

**02**

Predictive plans optimise sequence and baseline control; adaptive boards optimise flow and feedback; hybrid combines both intentionally.

**03**

Quality means conformance to requirements and fitness for use, not luxury or perfection.

**04**

Prevention and appraisal are usually cheaper than internal and external failure.

# Apply Methods and Tools to Drive a Project — Core Concepts II

**05**

A check sheet captures facts, a control chart detects process behaviour, Pareto prioritises causes, and fishbone plus 5 Whys investigates root cause.

**06**

Root-cause analysis must end at a process or system cause the team can influence, not blame a person.

**07**

Visual work management exposes blocked work and excessive WIP before the schedule report does.

**08**

Continuous improvement closes the loop through a named action, owner, measure and review date.

# Select the Tool for the Question

| Question | Best starting tool | Output |
|---|---|---|
| **What happened?** | Check sheet | Reliable facts |
| **Why might it happen?** | Fishbone | Cause families |
| **What is the root?** | 5 Whys | Actionable chain |
| **Which causes matter most?** | Pareto | Vital few |
| **Is the process stable?** | Control chart | Common vs special cause |

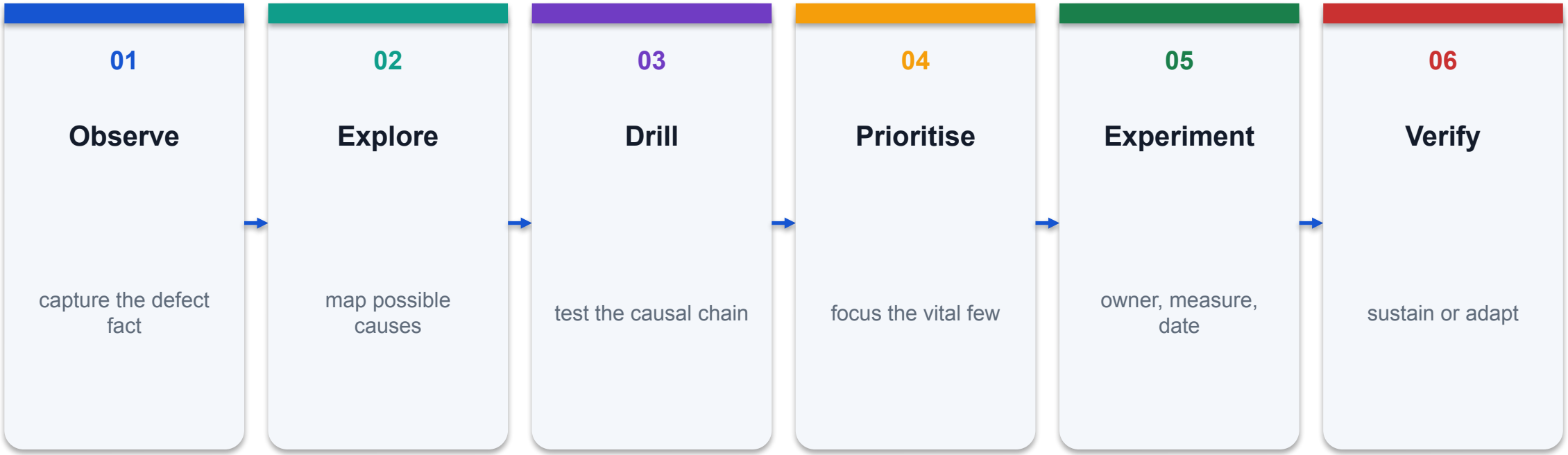
**Tools work as a sequence: facts → causes → priority → behaviour → action.**

# Cost of Quality

| Category | Purpose | Example |
|---|---|---|
| **Prevention** | Stop defects forming | Training and standards |
| **Appraisal** | Detect before release | Review and testing |
| **Internal failure** | Fix before customer | Rework and retest |
| **External failure** | Recover after release | Returns and reputation |

**Invest upstream where a dollar of prevention avoids many dollars of failure.**

# Root Cause to Improvement




**Stopping at a cause without a verified action is analysis theatre.**

# Method versus Tool

| Element | Purpose | Example |
|---|---|---|
| **Method** | Defines how work is managed | Predictive, Scrum, Kanban, PDCA |
| **Tool** | Makes evidence visible | WBS, board, chart, log |
| **Policy** | Defines operating rule | WIP limit, acceptance rule |
| **Cadence** | Defines review rhythm | Daily review, weekly status |

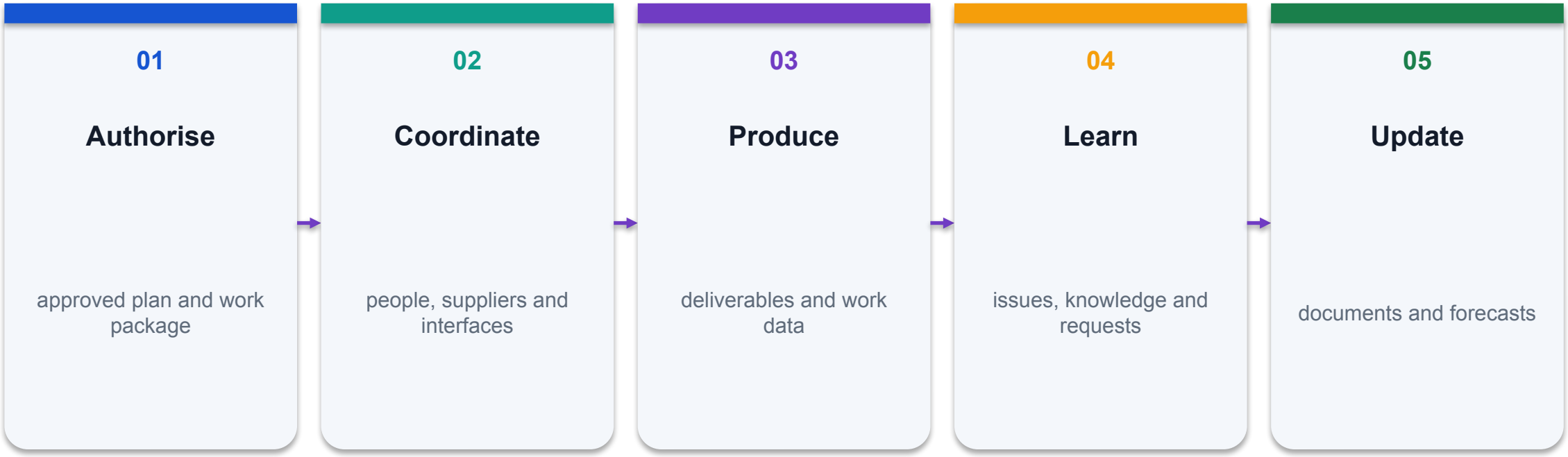
**A familiar tool used without a decision rule becomes decoration.**

# Predictive, Adaptive and Hybrid

| Signal | Predictive | Adaptive / hybrid |
|---|---|---|
| **Requirements** | Stable and knowable | Evolving or discovery-heavy |
| **Delivery** | Larger gated releases | Short usable increments |
| **Change** | Controlled against baseline | Expected inside prioritised backlog |
| **Best fit** | Low uncertainty, costly rework | Fast feedback, reversible choices |

**Tailor by deliverable when different parts of one small project carry different uncertainty.**

# Direct and Manage Project Work




**Execution produces both deliverables and information needed for control.**

# A Lean PMIS for a Small Project

### Source of truth

Named location and owner.

### Version control

Number, date, author and status.

### Decision log

What, who, when and rationale.

### Access

Right people, current artefact.

### Archive

Superseded records remain traceable.

### Security

Sensitive content has proportionate access.

# Quality Planning, Assurance and Control

| Discipline | Question | Evidence |
| --- | --- | --- |
| **Planning** | What quality is required? | Metrics, methods and acceptance |
| **Assurance** | Will the process achieve it? | Review, audit, coaching |
| **Control** | Does the result conform? | Inspection, tests and charts |
| **Improvement** | How will the system change? | Experiment and verification |

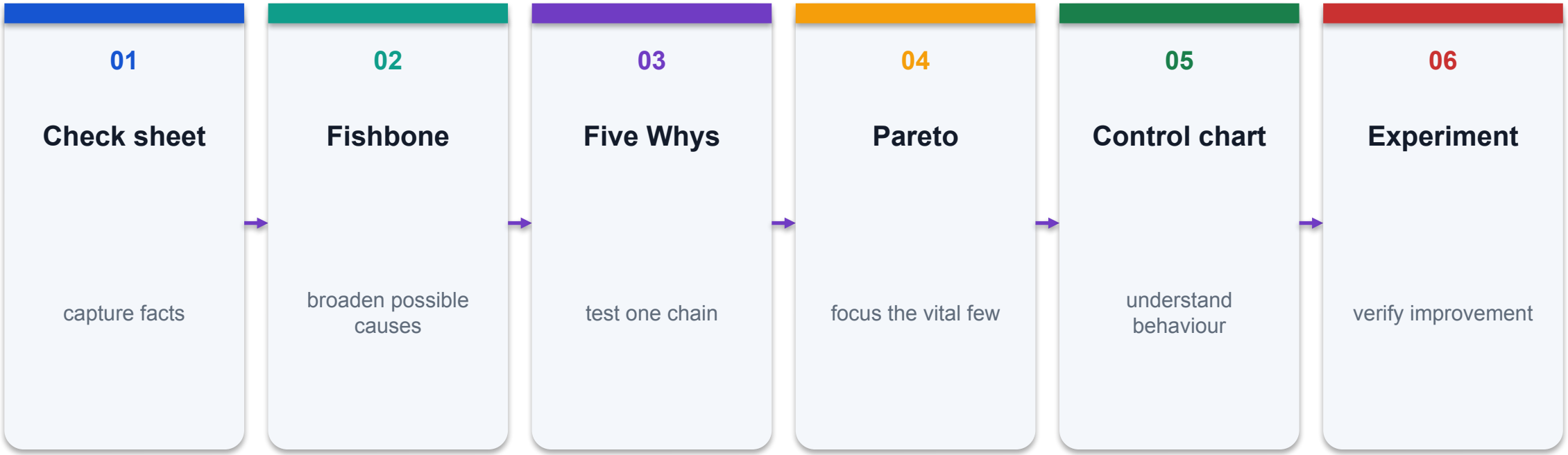
**Inspection finds defects; assurance and improvement reduce the chance of creating them.**

# HarbourHub Quality-Cost Decision

| Option | Near-term effect | Decision signal |
|---|---|---|
| **Add catalogue-data validation** | S$4,000 prevention spend | Reduces rejection and re-pick risk |
| **Test only at pilot launch** | Lower spend now | Defects discovered with customers waiting |
| **Accept manual correction** | No tooling change | Recurring labour and queue growth |
| **Decision** | Fund validation before rollout | Cheaper than repeated external failure |

**The team chooses prevention because one avoided launch-week failure can repay the control.**

# Quality Tool Sequence




**Do not jump from anecdote directly to solution.**

# Common Cause versus Special Cause

| Signal | Meaning | Response |
| --- | --- | --- |
| **Stable random variation** | System behaviour | Improve the process |
| **Point beyond limits** | Potential special cause | Investigate event |
| **Run or trend** | Non-random pattern | Test changed condition |
| **Within limits but off specification** | Stable but incapable | Redesign process |

**Control limits come from process data; specification limits come from requirements.**

# Kanban Operating Policies

### Visualise

Show status, owner and blockers.

### Limit WIP

Stop starting; start finishing.

### Manage flow

Watch ageing and bottlenecks.

### Make policies explicit

Entry, exit and priority rules.

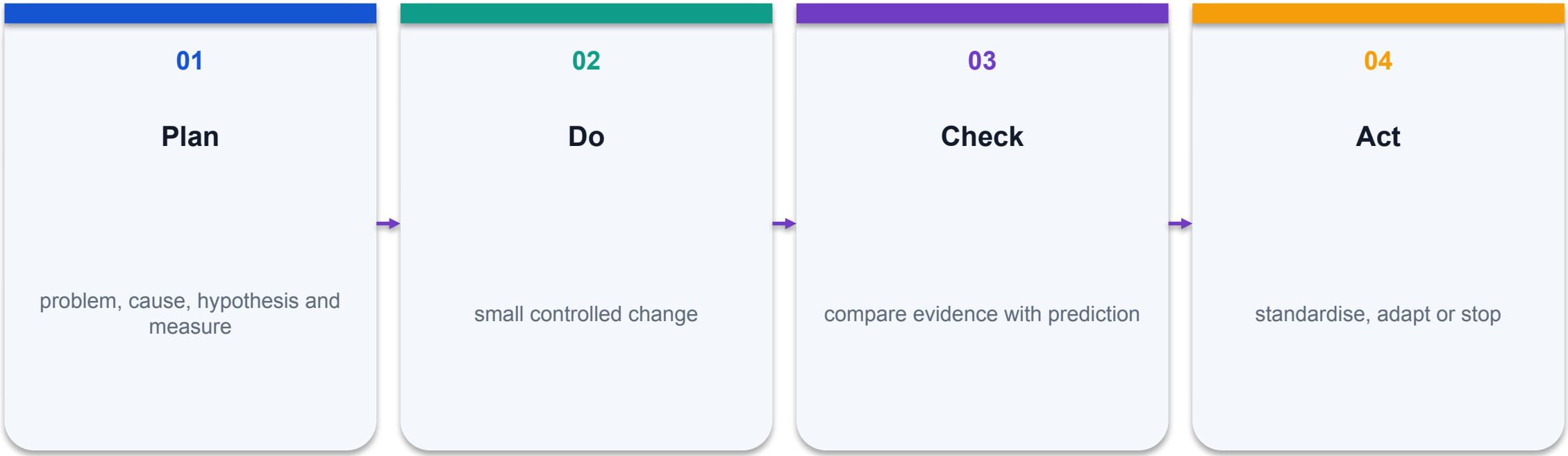
### Feedback

Review at a useful cadence.

### Improve

Run small measurable experiments.

# PDCA Improvement Loop




**Improvement is complete only when the new practice is verified and sustained.**

# Manage Project Knowledge

**Explicit**

Documents, data and procedures.

**Tacit**

Experience and judgement shared through work.

**Create**

Reviews, experiments and problem solving.

**Use**

Decision support during delivery.

**Transfer**

Handover and lessons for future teams.

# Activity 6 Tool Pack

### 30 Kanban Board

XLSX · editable project tool

### 35 Check Sheets

XLSX · editable project tool

### 34 Control Chart

XLSX · editable project tool

### 06 Delivery Approach and Quality Plan

XLSX · editable project tool

ACTIVITY 6 · 35 MIN

# Tailor the Delivery Method and Quality Plan

## THE SITUATION

The catalogue and POS interface need predictable milestone control, while the outlet workflow needs rapid testing and feedback. The team must choose a hybrid delivery rhythm and define quality before build begins.

## WHAT YOU'LL DO

Assess uncertainty and reversibility, tailor predictive and adaptive controls, define the Definition of Done, quality metrics, review cadence, WIP limits and evidence rules.

## YOU'LL PRODUCE

A documented hybrid approach, phase and iteration rhythm, quality management plan, Definition of Done, Kanban policy and measurement plan.

## DONE WHEN

The approach matches uncertainty, quality is measurable before work begins, WIP and evidence rules are explicit, and predictive/adaptive controls do not contradict one another.

**Tool: Delivery Approach Canvas, Quality Plan and Kanban Board**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 6 folder.

# Debrief — Activity 6: What It Proves

1

Hybrid is intentional tailoring, not running two incompatible systems in parallel.

2

Quality must be designed into the workflow before it is inspected at the end.

3

A tool earns its place only when it changes a decision or behaviour.

# Activity 10 Tool Pack

### 35 Check Sheets

XLSX · editable project tool

### 37 Ishikawa Diagram

XLSX · editable project tool

### 38 Five Whys

XLSX · editable project tool

### 36 Pareto Principle Template

XLSX · editable project tool

### 34 Control Chart

XLSX · editable project tool

ACTIVITY 10 · 40 MIN

# Diagnose the Pilot Quality Failure

## THE SITUATION

During testing, only 82% of orders are ready on time and pick accuracy is 94%. The vendor blames store staff; store staff blame inventory updates. The sponsor wants action before facts and hypotheses are separated.

## WHAT YOU'LL DO

Reconcile defect data, broaden possible causes, test a causal chain, prioritise the vital few, interpret process behaviour and design one measurable corrective experiment.

## YOU'LL PRODUCE

A check sheet, fishbone, five-whys chain, Pareto chart, control-chart interpretation and SMART corrective experiment.

## DONE WHEN

Data totals reconcile, Pareto reaches 100%, the causal chain is actionable, the control interpretation matches the signal and the experiment has an owner and verification date.

**Tool: Check Sheet, Fishbone, Five Whys, Pareto and Control Chart**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 10 folder.

# Debrief — Activity 10: What It Proves

1

Fishbone broadens, Five Whys deepens and Pareto prioritises.

2

Control limits describe process behaviour; specifications describe customer need.

3

Improvement is project work only when funded, owned and verified.

# Topic 03 Recap

### Decision

A method defines how work is managed; a tool makes information visible. Choose both for the decision needed.

### Method

Prevention and appraisal are usually cheaper than internal and external failure.

### Evidence

A check sheet, fishbone, five-whys chain, Pareto chart, control-chart interpretation and SMART corrective experiment.

### Capability

Use evidence and root-cause tools to select a controlled corrective experiment (A3, K3).

04

TOPIC 04 · A4, K4 · LO4

# Manage Project Risk

Identify · assess · respond · contingency · monitor

# Risk management buys options before they become expensive.

**The register matters only when it changes funding, sequencing, ownership or contingency before a trigger becomes an issue.**

# Manage Project Risk — Core Concepts I

**01**

Risk is an uncertain event or condition with a positive or negative effect; an issue has already happened.

**02**

Write risks as cause-event-impact statements so the response targets the cause or reduces the impact.

**03**

The risk register records owner, probability, impact, exposure, triggers, response, contingency and residual risk.

**04**

Qualitative analysis prioritises quickly; quantitative analysis estimates the range of outcomes and decision value.

# Manage Project Risk — Core Concepts II

**05**

Threat responses are avoid, mitigate, transfer, accept or escalate; opportunity responses are exploit, enhance, share, accept or escalate.

**06**

A contingency plan is triggered by an observable threshold and funded from contingency reserve.

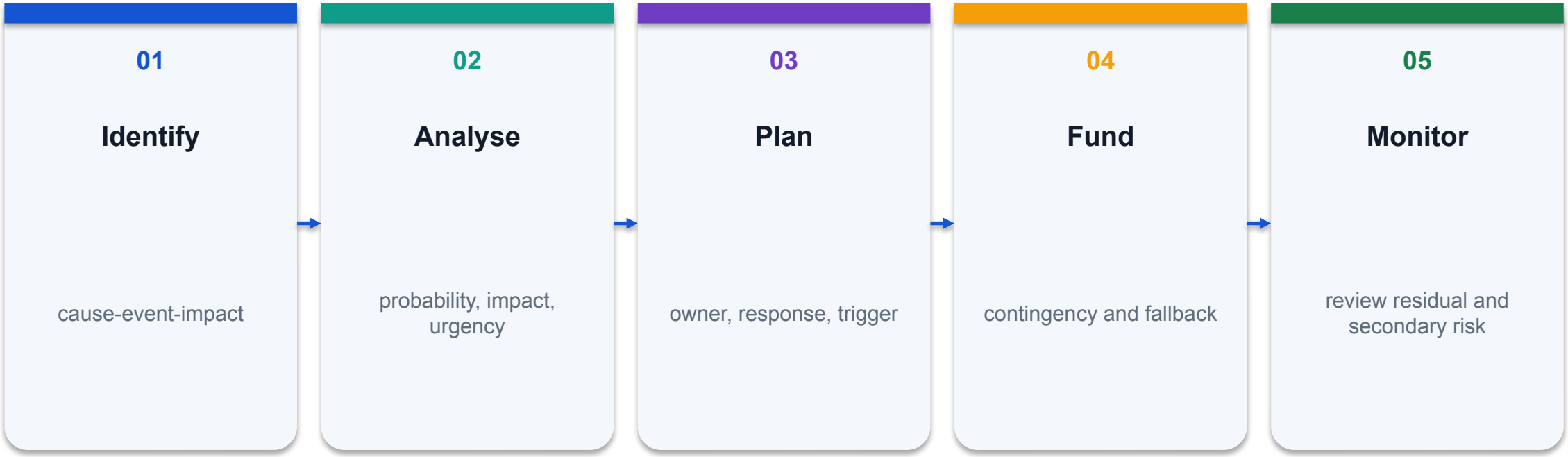
**07**

Decision trees compare expected monetary value; tornado charts reveal which assumptions deserve the most attention.

**08**

Risk reviews are recurring decisions, not a register created once and archived.

# The Risk Cycle




**Triggered risk becomes an issue and moves into action management.**

# Threat Response Choices

| Response | What changes | Example |
|---|---|---|
| **Avoid** | Remove the threat source | Change approach |
| **Mitigate** | Reduce probability or impact | Prototype early |
| **Transfer** | Shift financial consequence | Insurance / contract |
| **Accept** | Monitor and prepare | Contingency plan |
| **Escalate** | Move beyond authority | Sponsor decision |

**Acceptance is a decision with monitoring—not forgetting the risk.**

# Qualitative vs Quantitative Analysis

| Lens | Qualitative | Quantitative |
|---|---|---|
| **Speed** | Fast prioritisation | More data and effort |
| **Output** | Relative exposure | Range, EMV or sensitivity |
| **Use** | Most project risks | Material decision uncertainty |
| **Caution** | Scoring bias | False precision |

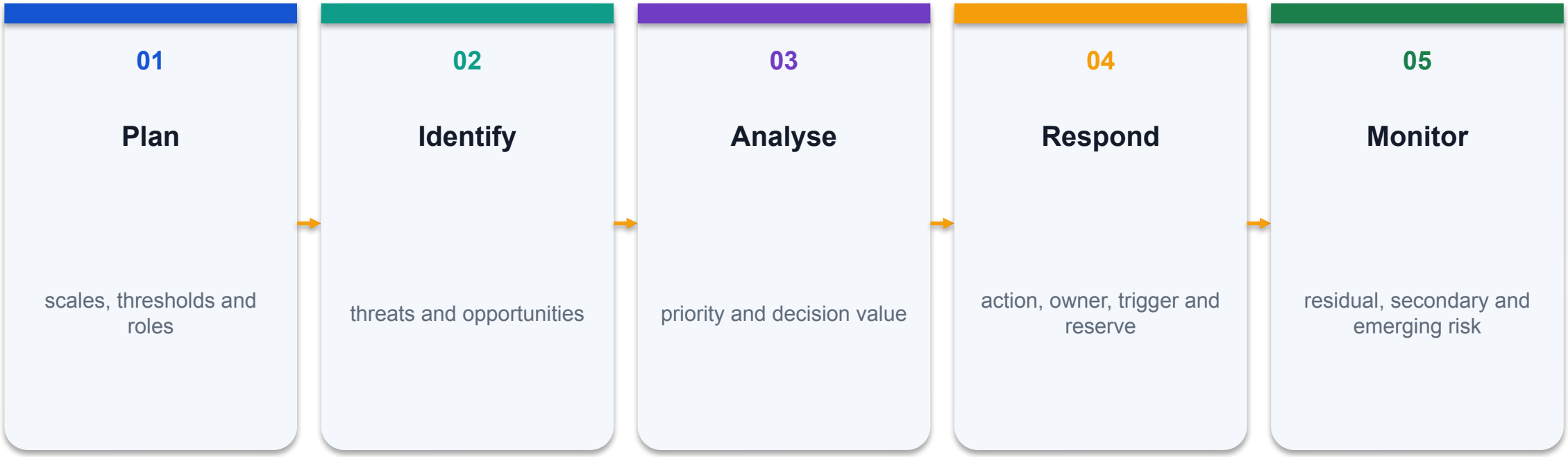
**Use quantitative analysis only where the decision value justifies the work.**

# Risk, Issue and Assumption

| Item | Time orientation | Management action |
|---|---|---|
| **Risk** | May happen | Owner, trigger and response |
| **Issue** | Has happened | Contain, resolve and escalate |
| **Assumption** | Believed true | Validate by a date |
| **Constraint** | Known limit | Plan within or change formally |

**A triggered risk becomes an issue; its response owner moves from readiness to action.**

# The Risk Management Cycle




**Risk management repeats whenever scope, schedule, suppliers or the environment change.**

# Risk Breakdown Structure

### Technical

Design, interface, data and quality.

### External

Regulation, market, weather and supplier.

### Organisational

Priority, funding, skills and governance.

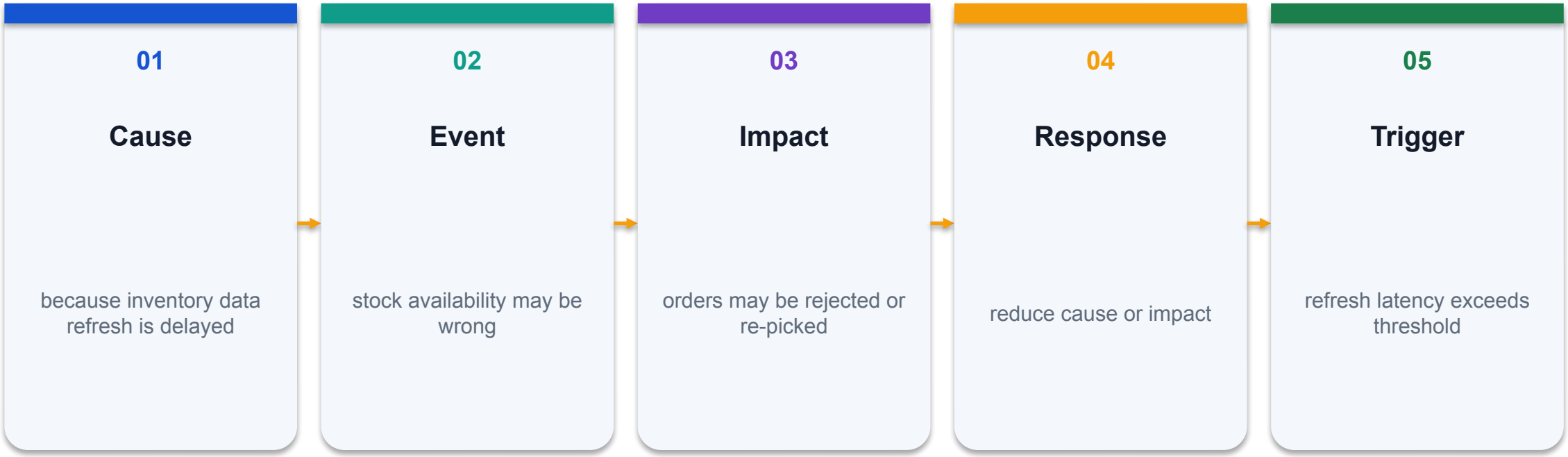
### Project management

Estimate, dependency, communication and scope.

### Opportunity

Faster adoption, reuse, cost saving and capability.

# Writing an Actionable Risk




**Cause-event-impact wording prevents vague risks such as 'vendor risk'.**

# Qualitative Risk Factors

| Factor | Question | Use |
|---|---|---|
| **Probability** | How likely? | Exposure priority |
| **Impact** | How severe? | Tolerance and reserve |
| **Urgency** | How soon must we act? | Response timing |
| **Proximity** | How soon may it occur? | Monitoring cadence |
| **Dormancy** | How long before impact appears? | Early detection |

**Two risks with the same score may require different action timing.**

# HarbourHub Threat-Response Workshop

| Named threat | Chosen response | Owned action and trigger |
|---|---|---|
| **Inventory feed is late** | Mitigate | IT lead adds latency alert; trigger >15 minutes |
| **Vendor misses integration date** | Transfer + mitigate | Milestone clause plus weekly interface test |
| **Outlet rejects new process** | Mitigate | Ops lead pilots job aid and observes first 20 orders |
| **Cyber approval exceeds tolerance** | Escalate | Sponsor decides scope trade by gate date |
| **Residual launch disruption** | Accept actively | Fund two-day floor support contingency |

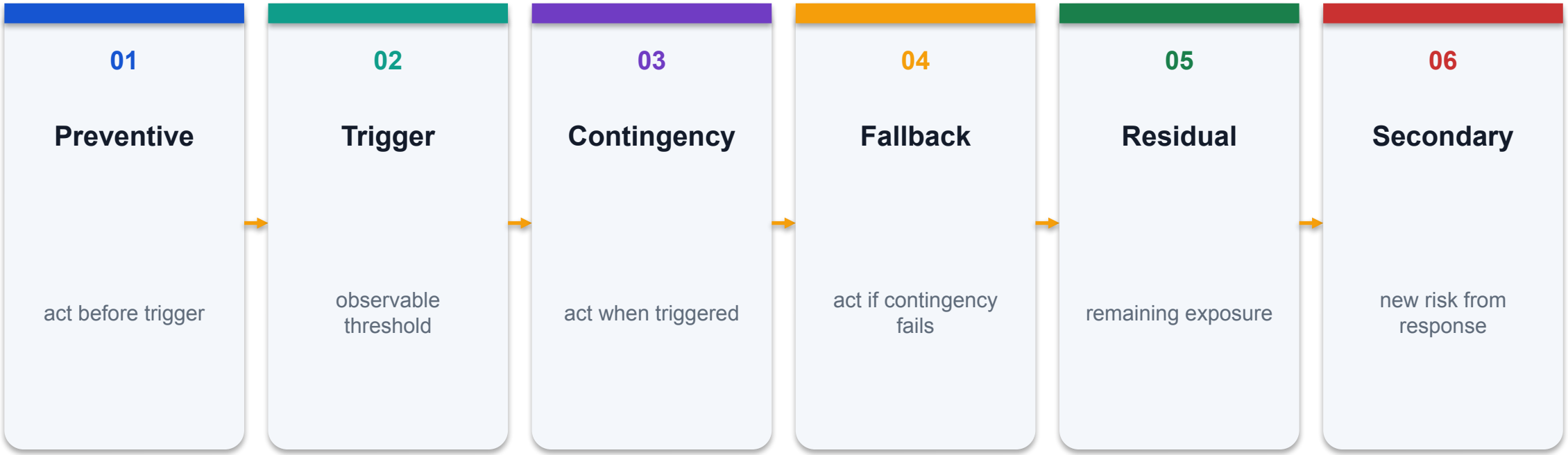
**Each response is specific enough to schedule, fund, monitor and verify.**

# Opportunity Response Strategies

| Strategy | Intent | Example |
|---|---|---|
| **Exploit** | Ensure opportunity occurs | Assign best resources |
| **Enhance** | Increase probability or benefit | Accelerate pilot learning |
| **Share** | Partner for capability | Revenue-sharing supplier |
| **Accept** | Take benefit if it occurs | No proactive spend |
| **Escalate** | Move to programme or strategy | Enterprise reuse |

**Positive uncertainty deserves ownership and deliberate action too.**

# Response Architecture




**A response is not operational until timing, ownership and funding are explicit.**

# Decision Tree and Sensitivity

| Tool | Question answered | Caution |
|---|---|---|
| **Decision tree** | Which option has better expected value? | Average value can hide tail consequences |
| **Tornado chart** | Which assumption drives outcome most? | Range quality determines insight |
| **Probability-impact matrix** | What needs attention first? | Ordinal scores are not precise money |

**Use more analysis only when the decision value exceeds the analysis effort.**

# Risk Review Agenda

### Triggered

Promote to issue and execute response.

### Changed

Re-score probability, impact and timing.

### New

Add emerging and secondary risk.

### Obsolete

Close with rationale.

### Reserve

Reconcile use and remaining exposure.

### Escalation

Decide beyond tolerance.

# Activity 7 Tool Pack

XLSX

**48 Risk Assessment Matrix Two pages**

Editable evidence file

XLSX

**45 Decision Tree Analysis TDIDT**

Editable evidence file

XLSX

**49 Tornado Chart - Sensitivity Analysis**

Editable evidence file

DECISION ENABLED

**A prioritised risk register, response and trigger plan, EMV recommendation, tornado sensitivity reading and reconciled S$18,000 contingency allocation.**

ACTIVITY 7 · 45 MIN

# Build the Risk Register and Fund Responses

## THE SITUATION

The POS interface may be late, refrigerated stock counts are unreliable, a key trainer has a short availability window and a faster payment option could improve adoption. Contingency reserve is S$18,000.

## WHAT YOU'LL DO

Write cause-event-impact risks, rank exposure, assign owners and triggers, choose responses, compare alternatives using EMV and sensitivity, and allocate contingency.

## YOU'LL PRODUCE

A prioritised risk register, response and trigger plan, EMV recommendation, tornado sensitivity reading and reconciled S$18,000 contingency allocation.

## DONE WHEN

Priority risks have owners, triggers and funded responses; the EMV and sensitivity logic reconcile; contingency is allocated to named uncertainty and residual risk remains visible.

**Tool: Risk Matrix, Decision Tree and Tornado Analysis**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 7 folder.

# Debrief — Activity 7: What It Proves

1

Risk management creates options before triggers become expensive issues.

2

Quantitative analysis sharpens material decisions but does not replace judgement.

3

Acceptance means informed monitoring and contingency, not forgetting the risk.

# Topic 04 Recap

### Decision

Risk is an uncertain event or condition with a positive or negative effect; an issue has already happened.

### Method

Qualitative analysis prioritises quickly; quantitative analysis estimates the range of outcomes and decision value.

### Evidence

A prioritised risk register, response and trigger plan, EMV recommendation, tornado sensitivity reading and reconciled S$18,000 contingency allocation.

### Capability

Identify, analyse and manage uncertainty with proportionate, funded action (A4, K4).

DAY 1 CLOSE

# End of Day 1

Save Activities 1–7. Day 2 continues the same project from the approved risk-aware baseline.

05

TOPIC 05 · A5, K5 · LO5

# Collaborate with Stakeholders

Identify · analyse · communicate · engage · allocate accountability

# Stakeholder engagement is part of delivery.

**Every unresolved expectation eventually appears as a delay, rejected deliverable, unapproved invoice or uncontrolled change.**

# Collaborate with Stakeholders — Core Concepts I

**01**

A stakeholder can affect, be affected by, or perceive themselves to be affected by the project.

**02**

The stakeholder register captures interest, influence, expectations, impact, communication needs and engagement strategy.

**03**

Power-interest mapping guides attention; the salience model adds legitimacy and urgency.

**04**

The engagement assessment matrix compares current and desired states so the gap becomes actionable.

# Collaborate with Stakeholders — Core Concepts II

**05**

RACI clarifies who is Responsible, Accountable, Consulted and Informed; every deliverable needs one clear Accountable owner.

**06**

Communication planning matches audience, message, channel, frequency, owner and feedback route.

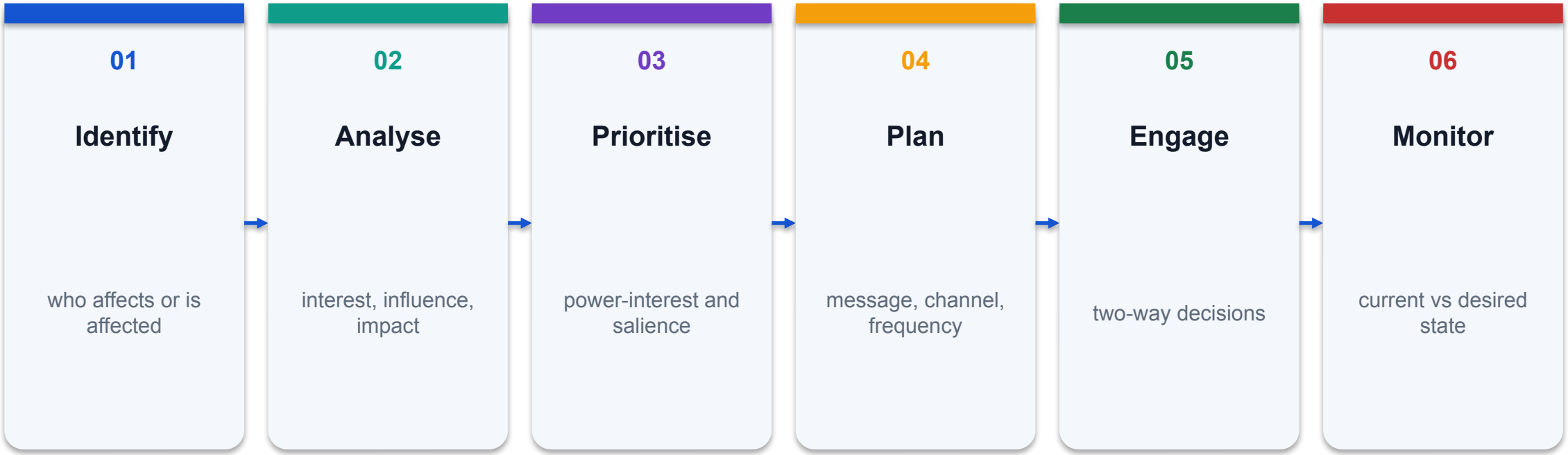
**07**

Engagement is two-way sense-making and decision participation, not one-way broadcasting.

**08**

Escalation should state the decision needed, options, impact of delay and recommendation.

# Stakeholder Analysis to Engagement




**A stakeholder register is useful only when it changes an engagement action.**

# RACI Diagnostic

| Pattern | Meaning | Correction |
|---|---|---|
| **No A** | Nobody owns the outcome | Assign one accountable owner |
| **Multiple A** | Competing decision rights | Choose one and define escalation |
| **No R** | No one performs the work | Assign delivery responsibility |
| **Too many C** | Decision gridlock | Consult only for needed input |
| **Everyone I** | Broadcast without purpose | Tailor information needs |

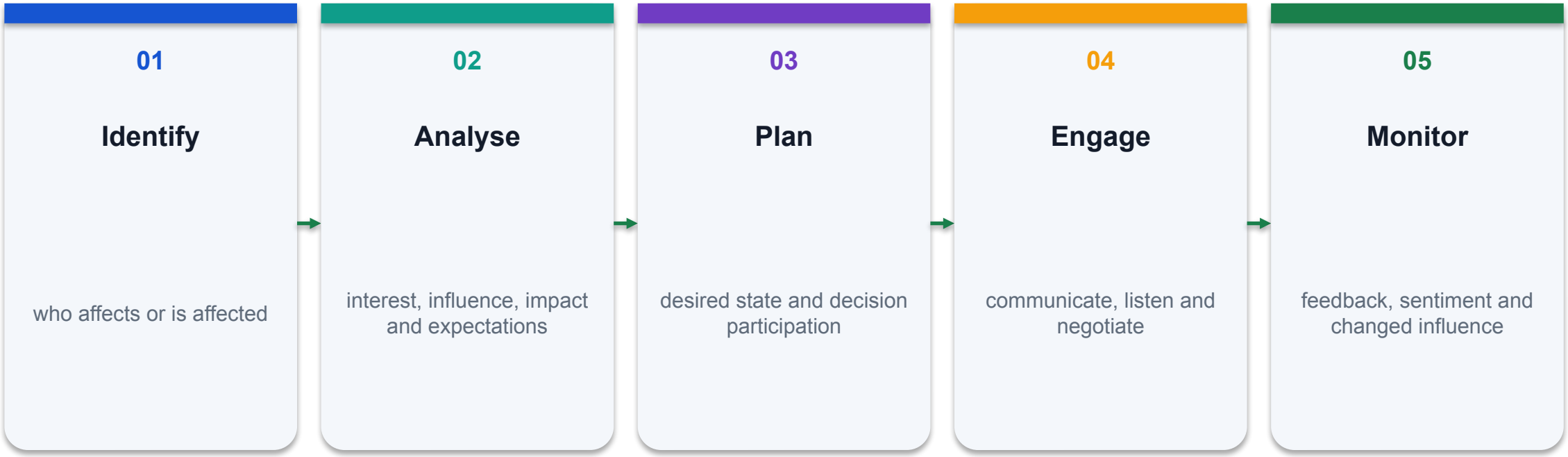
**Use deliverables and decisions as rows; vague verbs create vague accountability.**

# Communication vs Engagement

| Dimension | Communication | Engagement |
|---|---|---|
| **Direction** | Often one-way | Two-way |
| **Purpose** | Share information | Shape decisions and commitment |
| **Evidence** | Message delivered | Feedback changed the work |
| **Risk** | Noise and overload | Unresolved expectations |

**The channel matters less than whether the stakeholder can influence the decision in time.**

# The Stakeholder Engagement Cycle




**Stakeholder identification is continuous because project decisions change who is affected.**

# Stakeholder Register Fields

### Identity

Role, organisation and relationship.

### Interest

What outcome matters.

### Influence

Formal and informal ability to affect work.

### Impact

How the project changes their work.

### Expectations

Definition of success and concern.

### Strategy

Engagement action and owner.

# Stakeholder Mapping Lenses

| Lens | Dimensions | Best use |
|---|---|---|
| **Power-interest** | Authority and attention | Management strategy |
| **Impact-influence** | Effect and ability to act | Change and adoption |
| **Power-influence** | Authority and active influence | Governance |
| **Stakeholder cube** | Three dimensions | Complex populations |
| **Salience** | Power, legitimacy, urgency | Competing claims |

**A map is a hypothesis to guide action, not a permanent label.**

# Seven Salience Classes

**Dormant**

Power only.

**Discretionary**

Legitimacy only.

**Demanding**

Urgency only.

**Dominant**

Power and legitimacy.

**Dangerous**

Power and urgency.

**Dependent**

Legitimacy and urgency.

**Definitive**

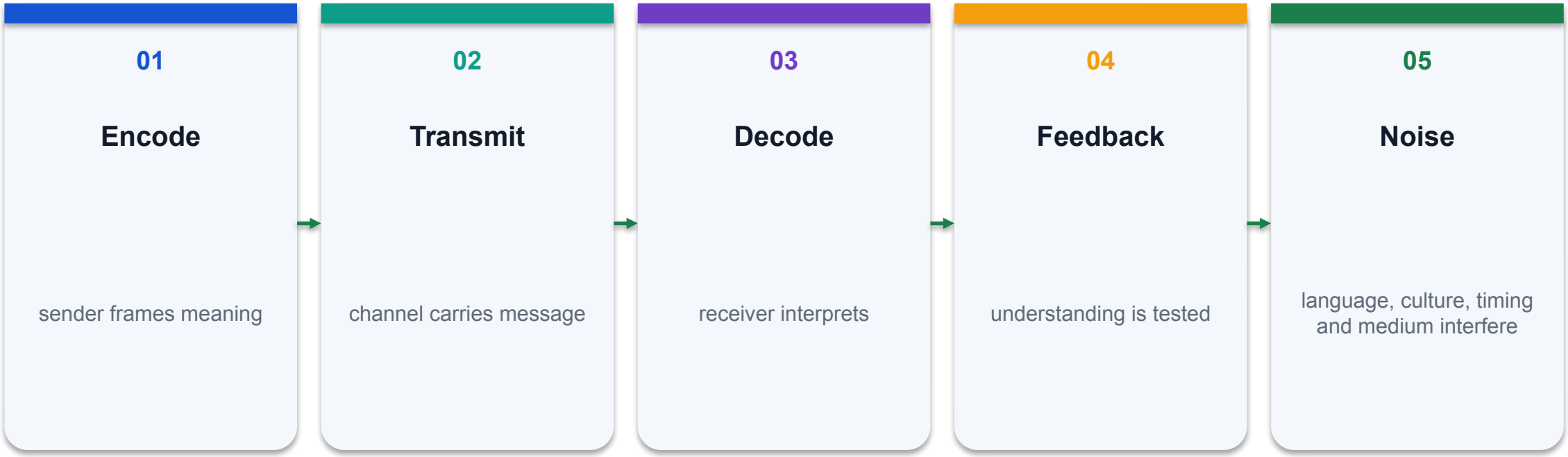
Power, legitimacy and urgency.

# Directions of Influence

| Direction | Typical stakeholders | Leadership move |
|---|---|---|
| **Upward** | Sponsor and executives | Decision brief and escalation |
| **Downward** | Team and specialists | Clarity, coaching and feedback |
| **Sideward** | Peers and functional managers | Negotiation and coordination |
| **Outward** | Customers, vendors, regulators | Expectation and interface management |

**Influence without authority depends on evidence, trust and a clear decision request.**

# Communication Model




**Delivery is not communication success; shared understanding is.**

# Communication Methods

| Method | Best use | Risk |
|---|---|---|
| **Interactive** | Decision and ambiguity | Time and facilitation |
| **Push** | Targeted distribution | Receipt is not understanding |
| **Pull** | Large reference set | Stakeholder may not retrieve it |
| **Visual radiator** | Shared current state | Stale information destroys trust |

**Choose channel based on urgency, complexity, sensitivity and need for feedback.**

# Stakeholder Engagement Levels

| Level | Signal | Management aim |
|---|---|---|
| **Unaware** | Does not know project | Build awareness |
| **Resistant** | Opposes change | Understand concern and involve |
| **Neutral** | Neither resists nor supports | Connect to relevant value |
| **Supportive** | Helps when asked | Sustain participation |
| **Leading** | Actively champions | Enable influence |

**The gap between current and desired state must become an owned action.**

# HarbourHub RACI Failure Test

| Observed row | Failure signal | Repair |
|---|---|---|
| **Pilot go-live** | Sponsor and PM both marked A | Sponsor is sole A; PM remains R |
| **Inventory mapping** | Vendor named R with no internal owner | IT lead becomes A; vendor remains R |
| **Outlet job aid** | No R assigned | Operations lead owns production |
| **Customer message** | Six people marked C | Consult legal and operations only |
| **Benefits review** | Everyone marked I | Name finance owner and decision audience |

**The repaired matrix makes every critical deliverable and decision executable.**

# Evidence-Based Escalation




**Escalation transfers a decision, not a problem without analysis.**

# Activity 8 Tool Pack

### 42 Stakeholder Impact Over Influence Matrix

XLSX · editable project tool

### 43 Salience Chart

PPTX · editable project tool

### 39 RACI Template

XLSX · editable project tool

### 41 Stakeholder Engagement Assessment Matrix

XLSX · editable project tool

ACTIVITY 8 · 40 MIN

# Align Stakeholders, Accountability and Communications

## THE SITUATION

The POS vendor and Operations each believe the other owns interface acceptance. Finance has not approved a payment milestone, store managers receive conflicting instructions and one influential supervisor resists the pilot.

## WHAT YOU'LL DO

Map influence and salience, diagnose engagement gaps, clarify RACI, design a two-way communication rhythm and prepare an evidence-based escalation brief.

## YOU'LL PRODUCE

A stakeholder map, salience reading, validated RACI, engagement assessment, communication actions and one-minute escalation recommendation.

## DONE WHEN

Critical decisions have one Accountable owner, material engagement gaps have two-way actions and the escalation asks the authorised person for a clear, time-bound decision.

**Tool: Stakeholder Map, Salience Model, RACI and Engagement Assessment**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 8 folder.

# Debrief — Activity 8: What It Proves

1

Many communication failures begin as unclear decision rights.

2

Engagement is participation and feedback, not message broadcasting.

3

Attention follows influence, impact and urgency—not title alone.

# Topic 05 Recap

### Decision

A stakeholder can affect, be affected by, or perceive themselves to be affected by the project.

### Method

The engagement assessment matrix compares current and desired states so the gap becomes actionable.

### Evidence

A stakeholder map, salience reading, validated RACI, engagement assessment, communication actions and one-minute escalation recommendation.

### Capability

Collaborate with stakeholders and clarify decision rights for execution (A5, K5).

TOPIC 06 · A6 · LO6

# Deploy Resources

People · budget · equipment · procurement · team performance

# Capacity—not headcount—delivers the plan.

**A seven-person team may have only three usable person-days at the constraint. Resource deployment is the discipline of matching skills and time to the critical work.**

# Deploy Resources — Core Concepts I

**01**

Resource planning links each work package to required skills, quantities, timing and availability.

**02**

A responsibility assignment is not a capacity plan; availability must be tested against concurrent work.

**03**

Resource levelling changes dates to resolve over-allocation; smoothing stays within available float.

**04**

Fast tracking overlaps work and raises risk; crashing adds cost to critical-path work and may not shorten the project.

# Deploy Resources — Core Concepts II

**05**

Procurement decisions consider make-or-buy, lead time, quality, interfaces, contract type and vendor capability.

**06**

Tuckman's stages require adaptive leadership: direct, coach, support and delegate as the team matures.

**07**

Conflict is normal; collaborate for important relationships and outcomes, compromise when time is limited, and escalate only with evidence.

**08**

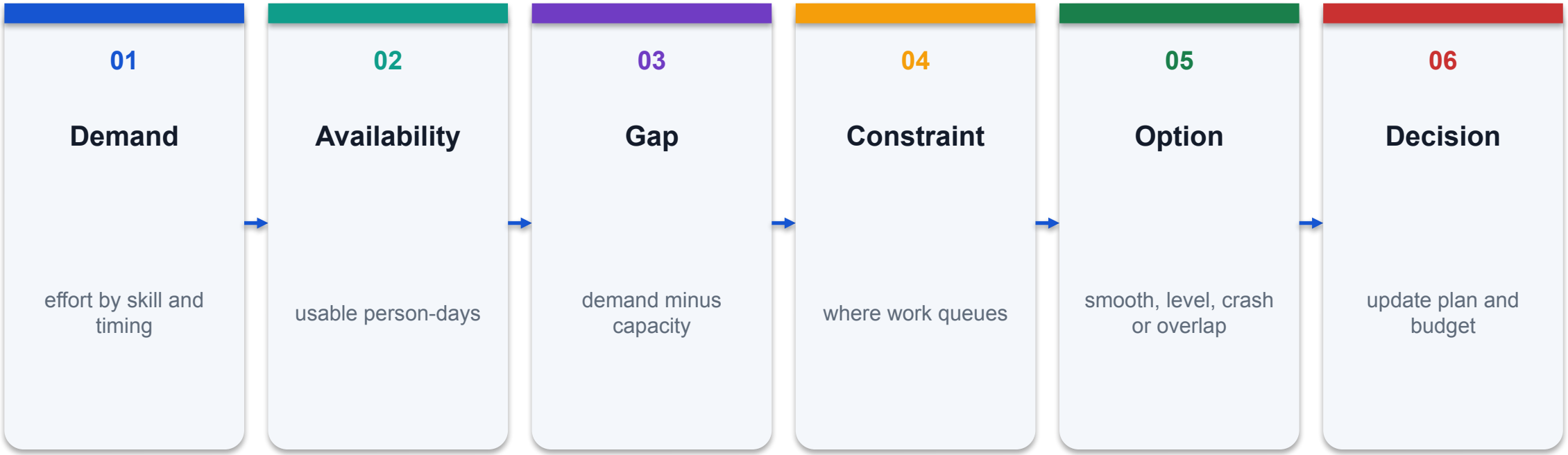
A resource dashboard should show demand, capacity, variance, bottlenecks and decisions—not just headcount.

# Four Resource Responses

| Option | Schedule effect | Cost / risk effect |
|---|---|---|
| **Smoothing** | Uses float; finish unchanged | Low cost, limited relief |
| **Levelling** | May delay finish | Resolves over-allocation |
| **Crashing** | Shortens critical work | Adds cost; diminishing returns |
| **Fast tracking** | Overlaps sequence | Adds rework and quality risk |

**Apply compression only to the current critical path and reanalyse after the change.**

# Capacity Planning




**Capacity is time and skill together; headcount alone is not a usable measure.**

# Team Stage and Leadership

| Stage | Team signal | Leader response |
|---|---|---|
| **Forming** | Unclear roles and dependence | Direct and clarify |
| **Storming** | Conflict and boundary testing | Coach and facilitate |
| **Norming** | Shared standards emerge | Support and reinforce |
| **Performing** | Autonomous delivery | Delegate and protect |
| **Adjourning** | Transition and loss | Recognise and close |

**Using one leadership style for every stage creates either dependence or frustration.**

# Resource Management Plan

### Identify

People, equipment, facilities and funding.

### Estimate

Quantity, skill and timing.

### Acquire

Internal assignment or procurement.

### Develop

Capability, trust and working agreements.

### Manage

Performance, conflict and feedback.

### Control

Usage, availability and corrective action.

# Organisation Structure and PM Authority

| Structure | PM authority | Resource pattern |
|---|---|---|
| **Functional** | Low | Functional manager controls |
| **Weak matrix** | Limited | Coordinator role |
| **Balanced matrix** | Shared | Negotiated authority |
| **Strong matrix** | Moderate to high | Dedicated PM authority |
| **Project-oriented** | High | Dedicated team |

**Even a small project needs explicit escalation when formal authority is limited.**

# Role, Authority, Responsibility and Competence

| Element | Question | Evidence |
|---|---|---|
| **Role** | What function is performed? | Role description |
| **Authority** | What decisions may be made? | Charter and tolerance |
| **Responsibility** | What outcome or work is owned? | RACI and work package |
| **Competence** | Can the person perform it? | Skill and experience |

**Assigning work without authority or competence creates hidden dependency.**

# Two-Week Capacity Worked Example

| Skill | Demand | Usable capacity / decision |
|---|---|---|
| **Integration developer** | 96 hours | 64 hours — defer reporting interface |
| **Outlet trainer** | 40 hours | 48 hours — retain buffer |
| **Data analyst** | 52 hours | 32 hours — buy 20 specialist hours |
| **Test lead** | 60 hours | 56 hours — smooth four hours into next week |

**The constraint is skill-specific usable time, not the team's nominal headcount.**

# Acquire, Develop or Procure

| Choice | Use when | Trade-off |
|---|---|---|
| **Internal assignment** | Capability exists | Priority conflict |
| **Develop skill** | Need is recurring | Learning time |
| **Contract resource** | Short-term specialist gap | Cost and knowledge transfer |
| **Outsource deliverable** | Clear result and interface | Control and vendor dependency |

**Choose the smallest intervention that resolves the critical constraint.**

# Make-or-Buy Decision

| Criterion | Make | Buy |
|---|---|---|
| **Capability** | Builds internal skill | Accesses specialist skill |
| **Lead time** | Depends on availability | Depends on procurement and onboarding |
| **Control** | Direct management | Contract and interface control |
| **Cost** | Fixed or opportunity cost | Price plus vendor management |
| **Knowledge** | Retained internally | Requires transfer plan |

**Lowest price is not lowest total project cost when delay and interface risk are included.**

# Contract Type and Risk

| Type | Cost risk | Best fit |
| --- | --- | --- |
| **Firm fixed price** | More seller risk | Stable, clear scope |
| **Cost reimbursable** | More buyer risk | Uncertain scope with control |
| **Time and materials** | Shared / capped | Short-term flexible effort |
| **Incentive arrangement** | Shared performance target | Measurable outcome |

**The contract allocates financial risk; it does not eliminate delivery accountability.**

# HarbourHub Team-Stage Diagnostic

| Observed behaviour | Stage diagnosis | Next leadership move |
|---|---|---|
| **Vendor and IT dispute interface ownership** | Storming | Facilitate roles and decision rule |
| **Outlet champions share one job aid** | Norming | Reinforce standard and peer coaching |
| **Test team resolves defects without escalation** | Performing | Delegate within tolerance |
| **Pilot staff worry about release** | Adjourning | Recognise contribution and confirm transition |

**Leadership changes with team evidence; it is not a fixed personal style.**

# Conflict Approaches

| Approach | When useful | Trade-off |
|---|---|---|
| **Collaborate** | Outcome and relationship both matter | Time and openness |
| **Compromise** | Time-limited middle ground | Partial satisfaction |
| **Accommodate** | Relationship or learning matters more | Your need deferred |
| **Direct / force** | Emergency or authority decision | Commitment risk |
| **Avoid** | Low value or cooling period | Issue may persist |

**Conflict is information; choose the response based on stakes and relationship.**

# HarbourHub Bottleneck Options

| Option tested | Schedule effect | Decision |
|---|---|---|
| **Smooth outlet training** | Uses two days of float | Accept — no milestone impact |
| **Level integration work** | Moves launch by one week | Reject unless scope is traded |
| **Crash data validation** | Recovers three days for S$3,600 | Approve inside reserve |
| **Fast-track testing** | May recover four days | Reject — high rework and quality risk |

**The selected combination protects launch without hiding cost or quality trade-offs.**

# Team Performance Signals

### Delivery

Commitments met with acceptance evidence.

### Flow

WIP, blockers and ageing visible.

### Quality

Defects and rework trend.

### Collaboration

Handoffs, feedback and conflict resolved.

### Learning

Capability and knowledge transfer.

### Sustainability

Workload remains safe and realistic.

# Activity 9 Tool Pack

### 40 Organisation Chart

PPTX · editable project tool

### 30 Kanban Board

XLSX · editable project tool

### 32 Project Budget - Planned vs Actual - Weekly and Monthly

XLSX · editable project tool

### 09 Resource Capacity and Procurement Decision

XLSX · editable project tool

ACTIVITY 9 · 45 MIN

# Deploy Resources and Resolve the Bottleneck

## THE SITUATION

Testing needs 22 person-days in two weeks, but only 16 are available. The business analyst is assigned to testing, training and vendor clarification. The sponsor wants evidence before accepting delay or extra cost.

## WHAT YOU'LL DO

Map authority and skills, calculate demand-capacity gaps, test smoothing, levelling, crashing and fast tracking, compare make/buy options and recommend a funded deployment decision.

## YOU'LL PRODUCE

An organisation map, capacity heatmap, constraint-aware Kanban, procurement comparison, compression options, budget impact and sponsor recommendation.

## DONE WHEN

Capacity reconciles to named work and skills, the bottleneck is explicit, procurement/compression options show complete trade-offs and the chosen deployment remains within authorised constraints.

**Tool: Organisation Chart, Resource Capacity, Procurement Decision, Kanban and Budget**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 9 folder.

# Debrief — Activity 9: What It Proves

1

A RACI assignment does not create capacity.

2

Manage the constraint before adding capacity elsewhere.

3

Crashing buys time with cost; fast tracking buys time with risk.

# Topic 06 Recap

### Decision

Resource planning links each work package to required skills, quantities, timing and availability.

### Method

Fast tracking overlaps work and raises risk; crashing adds cost to critical-path work and may not shorten the project.

### Evidence

An organisation map, capacity heatmap, constraint-aware Kanban, procurement comparison, compression options, budget impact and sponsor recommendation.

### Capability

Match skills, capacity, procurement and budget to critical work (A6).

TOPIC 07 · A7 · LO7

# Track Project Deliverables and Correct Misalignments

Performance · earned value · change · corrective action · handover · closure

# Control means deciding early enough to change the outcome.

**Variance reporting without forecast, ownership and action is history. Good control protects value and makes closure evidence-ready.**

# Track Project Deliverables and Correct Misalignments — Core Concepts I

**01**

Work performance data becomes information only after comparison with the approved baseline and thresholds.

**02**

A status report should answer: where are we, why, what is forecast, what decision is needed, and who owns the action.

**03**

Earned value integrates scope, schedule and cost: PV is planned work, EV is completed value, and AC is actual spend.

**04**

CPI = EV/AC and SPI = EV/PV; indices below 1.00 signal inefficiency against the baseline.

# Track Project Deliverables and Correct Misalignments — Core Concepts II

**05**

A change request states reason, impact, options and recommendation; approved changes update baselines and traceability.

**06**

Corrective action brings performance back to plan; preventive action reduces the probability of future variance.

**07**

Schedule compression is applied to the current critical path and re-analysed after every change.

**08**

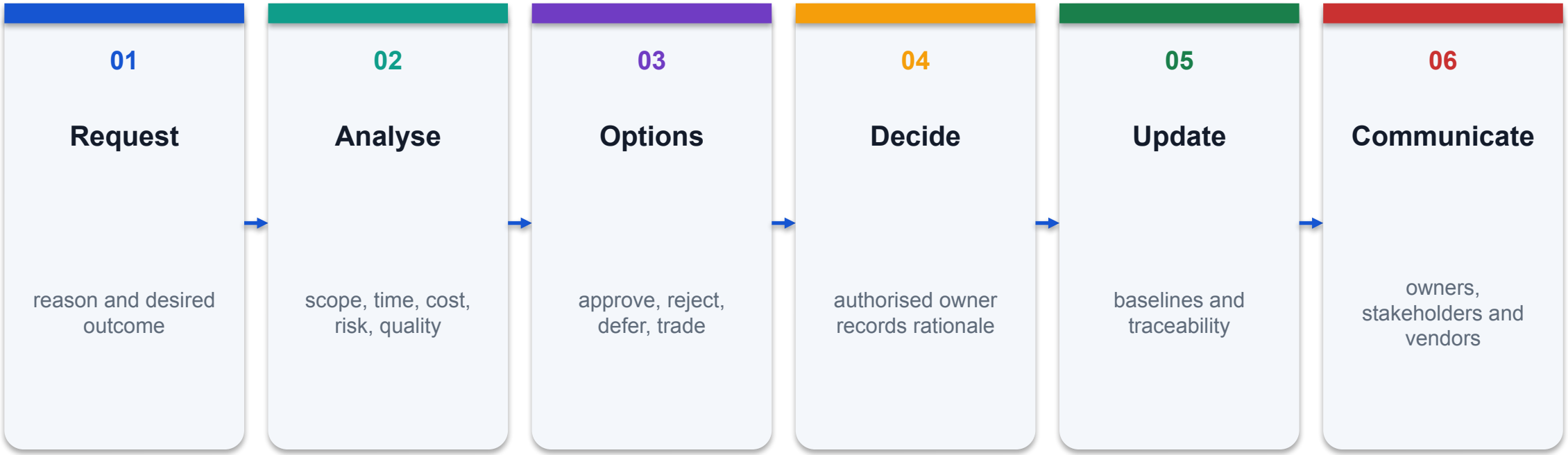
Closure confirms acceptance, transition, contract completion, records, benefits ownership and lessons learned.

# Earned Value at Week 12

| Measure | Formula / value | Meaning |
|---|---|---|
| **PV** | S$108,000 | Budgeted work planned |
| **EV** | S$90,000 | Budgeted value completed |
| **AC** | S$105,000 | Actual cost incurred |
| **SPI** | EV/PV = 0.83 | Behind schedule |
| **CPI** | EV/AC = 0.86 | Cost inefficient |

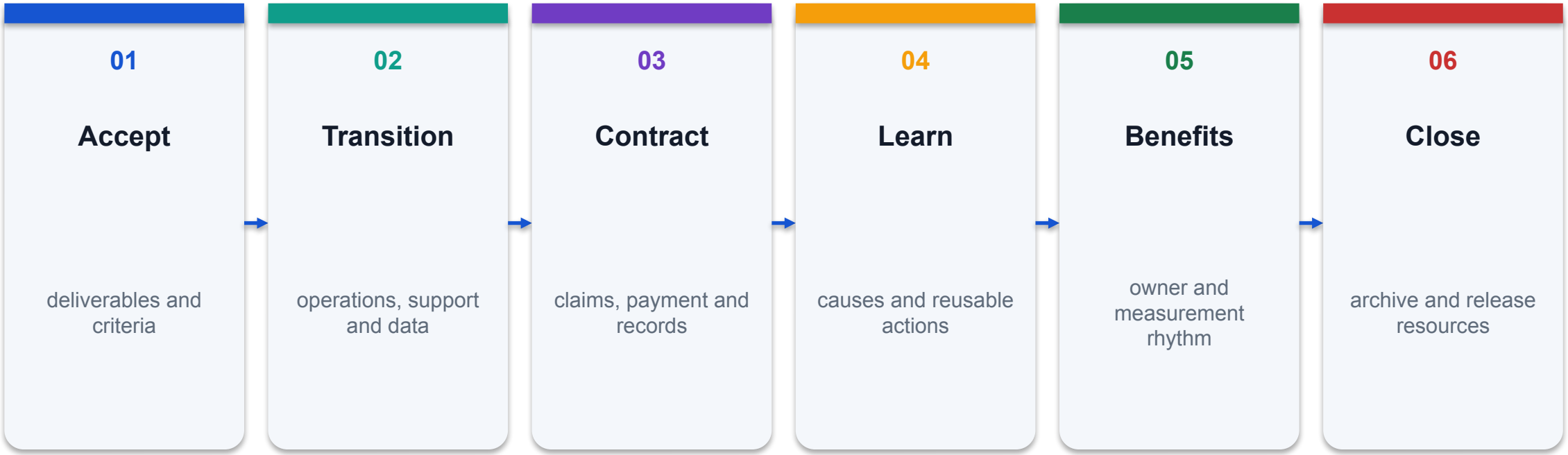
**Use the indices with milestones and quality evidence—no single metric tells the whole story.**

# Integrated Change Control



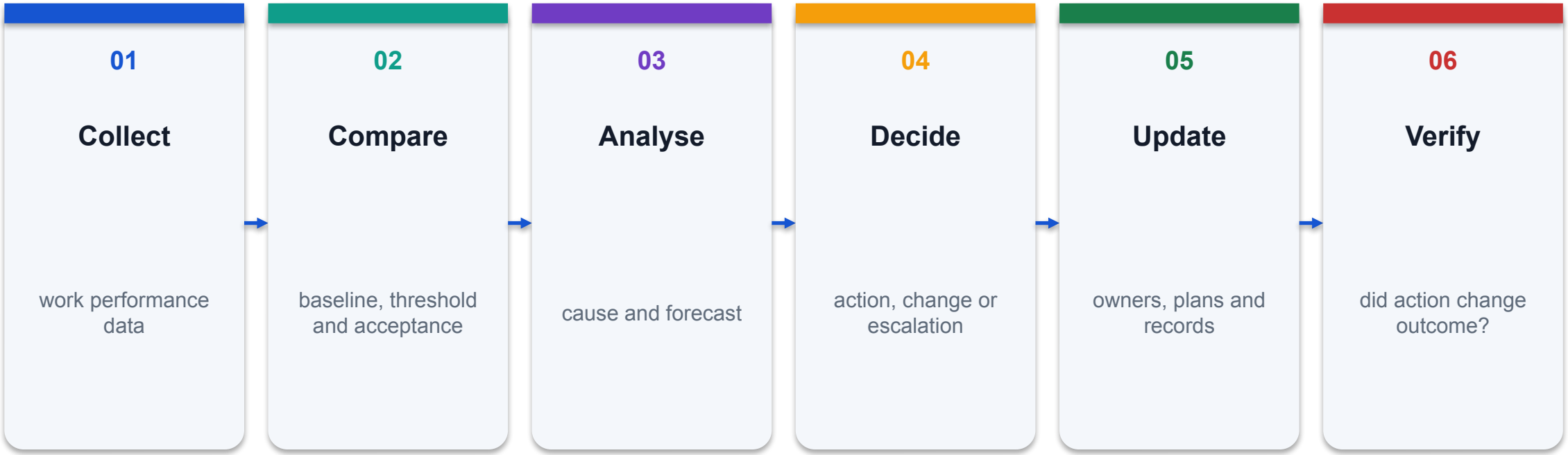

**An approved change is unfinished until affected baselines and acceptance evidence are updated.**

# Closeout Gates




**Do not close on elapsed time; close on complete evidence and transferred accountability.**

# The Monitoring and Control Cycle




**Control is forward-looking: the purpose of variance is to improve the forecast and decision.**

# Data, Information and Reports

| Level | Example | Use |
|---|---|---|
| **Data** | 7 defects, S$105k actual cost | Raw observation |
| **Information** | Defects above threshold, CPI 0.86 | Meaning through comparison |
| **Report** | Forecast, cause, action and decision | Governance and coordination |

**A dashboard without interpretation is data display, not project control.**

# A Decision-Ready Status Report

### Where

Current milestone and acceptance state.

### Variance

Scope, time, cost, quality and resource.

### Why

Cause, not just symptom.

### Forecast

Expected finish and cost.

### Action

Owner, due date and verification.

### Decision

Escalation with options and deadline.

# Validate Scope versus Control Scope

| Process | Question | Evidence |
|---|---|---|
| **Validate scope** | Will the authorised stakeholder accept it? | Accepted deliverable |
| **Control scope** | Has authorised scope changed? | Baseline, RTM and change log |
| **Control quality** | Does the result conform? | Inspection and test record |

**Verified quality is not the same as formal customer acceptance.**

# Earned Value Building Blocks

**PV**

Budgeted value of work planned by status date.

**EV**

Budgeted value of work actually completed.

**AC**

Actual cost of completed work.

**BAC**

Total authorised budget for measured scope.

**Status date**

The same cut-off for every measure.

# Variance and Performance Indices

| Measure | Formula | Interpretation |
|---|---|---|
| **SV** | EV − PV | Negative means less value completed than planned |
| **CV** | EV − AC | Negative means completed value cost more than budgeted |
| **SPI** | EV ÷ PV | Below 1.00 means schedule inefficiency |
| **CPI** | EV ÷ AC | Below 1.00 means cost inefficiency |

**Translate every index into operational consequence and required decision.**

# Forecasting with EAC

| Situation | Formula | Assumption |
|---|---|---|
| **Current cost efficiency continues** | EAC = BAC ÷ CPI | Past cost performance persists |
| **Remaining work re-estimated** | EAC = AC + bottom-up ETC | New detailed forecast is credible |
| **Original variance is one-off** | EAC = AC + (BAC − EV) | Future returns to plan |
| **Cost and schedule both constrain** | EAC = AC + (BAC − EV)/(CPI × SPI) | Both inefficiencies persist |

**Forecast choice is a management assumption and must be stated.**

# Issue Management




**Do not keep an issue disguised as a risk after it has occurred.**

# Loyalty-Points Change Worked Example

| Analysis lens | Finding | Implication |
|---|---|---|
| **Value** | Could lift repeat purchase | Benefit is plausible but unvalidated |
| **Scope and schedule** | Adds API and two test cycles | Launch slips nine working days |
| **Cost and resource** | S$18,000 plus scarce developer | Exceeds PM tolerance |
| **Risk and quality** | New data/privacy interface | Needs security and regression evidence |
| **Decision** | Defer to post-pilot release | Protect baseline; retain request and rationale |

**Useful ideas are not automatically urgent: the sponsor protects the pilot while preserving the option.**

# Three Types of Action

| Action | Purpose | Example |
|---|---|---|
| **Corrective** | Bring future performance back to plan | Add test capacity to recover milestone |
| **Preventive** | Reduce probability of future variance | Add interface review before integration |
| **Defect repair** | Modify non-conforming deliverable | Correct catalogue mapping |

**Each action needs authority, ownership, due date and verification.**

# Control Across Knowledge Areas

### Schedule

Critical path, float and milestone forecast.

### Cost

Actuals, earned value and forecast.

### Quality

Conformance, variation and corrective action.

### Resources

Utilisation, capacity and bottlenecks.

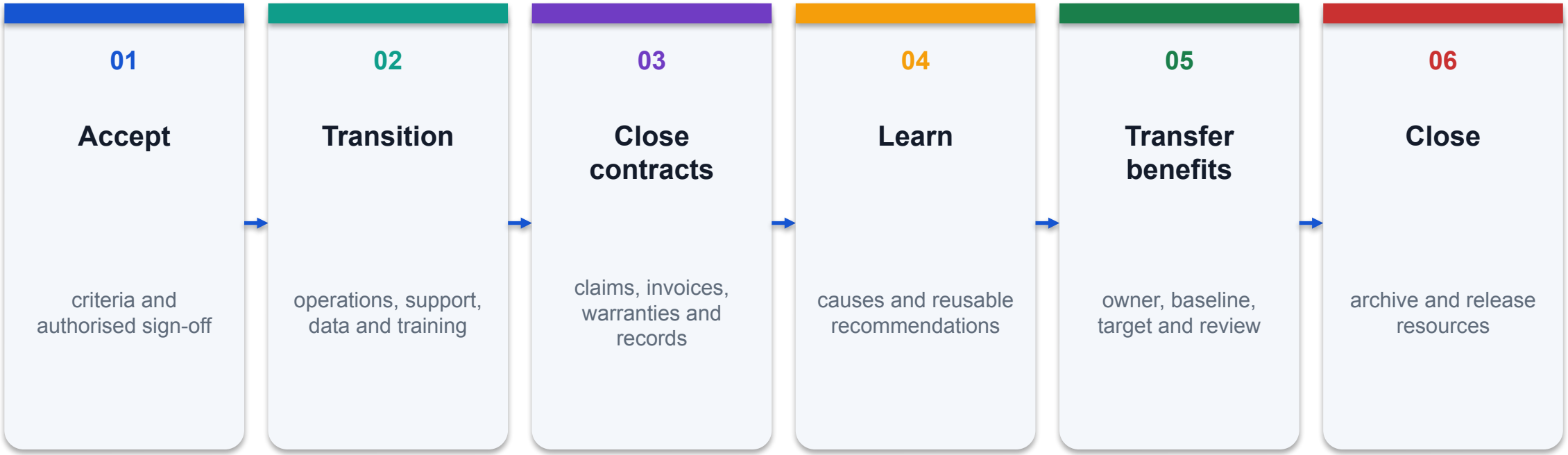
### Risk

Triggers, exposure and reserve.

### Communication & procurement

Effectiveness, obligations and disputes.

# Closure from Acceptance to Benefits




**The project can close only after unresolved work is completed or formally transferred.**

# Activity 11 Tool Pack

### 31 Burndown Chart

XLSX · editable project tool

### 32 Project Budget - Planned vs Actual - Weekly and Monthly

XLSX · editable project tool

### 34 Control Chart

XLSX · editable project tool

### 11 Integrated Status Change and Issue Log

XLSX · editable project tool

ACTIVITY 11 · 45 MIN

# Control Performance, Issues and Change

## THE SITUATION

At week 12, PV is S$108,000, EV is S$90,000 and AC is S$105,000. The sponsor requests loyalty points for S$24,000 and three weeks while a vendor interface issue threatens acceptance.

## WHAT YOU'LL DO

Calculate earned value and forecast, integrate quality and milestone evidence, triage the issue, analyse the change, obtain an authorised decision and define corrective action.

## YOU'LL PRODUCE

An integrated status report, EVM calculations and forecast, issue decision, change-impact analysis, decision log, corrective action and updated control records.

## DONE WHEN

EVM and forecast reconcile, issue and change records are distinct, the decision is authorised, all affected records are updated and corrective action is measurable.

**Tool: Integrated Status, Change and Issue Log, Burndown, Budget and Control Chart**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 11 folder.

# Debrief — Activity 11: What It Proves

1

Status reporting is a decision system, not a colour-coded history lesson.

2

Change control enables deliberate trade-offs; it does not prevent change.

3

An approved change is incomplete until every affected baseline is updated.

# Activity 12 Tool Pack

XLSX

**44 Agile Retrospective**

Editable evidence file

XLSX

**12 Acceptance Handover Closure and Benefits**

Editable evidence file

**DECISION ENABLED**

**Signed acceptance matrix, handover and training evidence, unresolved-item disposition, contract closeout, lessons register, benefits tracker and formal closure recommendation.**

ACTIVITY 12 · 35 MIN

# Accept, Handover and Close the Project

## THE SITUATION

All three outlets are live, but final payment is requested before complete acceptance evidence, support ownership, training records and unresolved-item decisions are consolidated. Benefits will continue after project closure.

## WHAT YOU'LL DO

Validate acceptance, transfer operations and knowledge, resolve contracts and open items, capture lessons and assign post-project benefits measures before recommending closure.

## YOU'LL PRODUCE

Signed acceptance matrix, handover and training evidence, unresolved-item disposition, contract closeout, lessons register, benefits tracker and formal closure recommendation.

## DONE WHEN

Acceptance is traceable, handover ownership is explicit, open items have formal disposition, contracts and records reconcile, and benefits continue under named operational owners.

**Tool: Acceptance, Handover, Closure and Benefits Tracker plus Retrospective**

Full procedure, evidence fields and acceptance criteria: Learner Guide and Activity 12 folder.

# Debrief — Activity 12: What It Proves

1

Elapsed time does not close a project; complete evidence and transferred accountability do.

2

Lessons become assets only when future owners can find and use them.

3

Project work ends before benefits measurement ends.

# Topic 07 Recap

### Decision

Work performance data becomes information only after comparison with the approved baseline and thresholds.

### Method

CPI = EV/AC and SPI = EV/PV; indices below 1.00 signal inefficiency against the baseline.

### Evidence

Signed acceptance matrix, handover and training evidence, unresolved-item disposition, contract closeout, lessons register, benefits tracker and formal closure recommendation.

### Capability

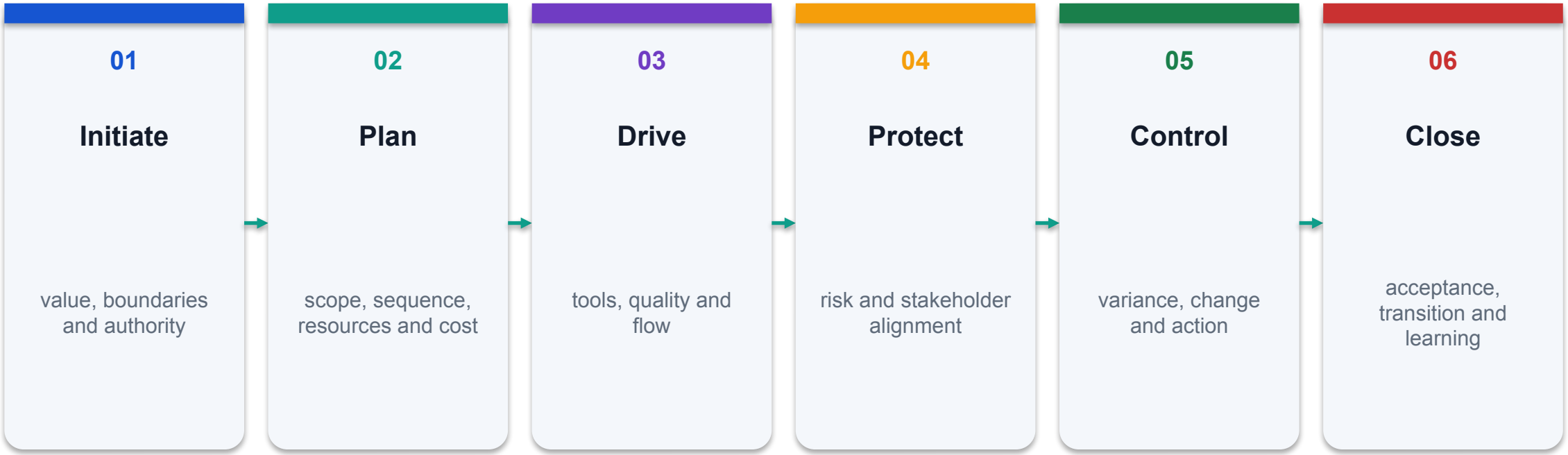
Close only when deliverables, transition, contracts, records and benefits ownership are evidenced (A7).

WRAP-UP

# Course Wrap-Up

From authorisation to controlled handover—one connected evidence chain.

# The Small-Project Delivery System




**Every output becomes an input to the next decision.**

# Lifecycle Takeaways

**01**

Start with value and measurable success, not a list of tasks.

**02**

Integrate scope, schedule and cost around the same work packages.

**03**

Choose the tool that changes the next decision and retire unused reporting.

**04**

Fund risk responses before triggers become issues.

**05**

Make accountability and stakeholder feedback visible.

**06**

Plan capacity at the constraint, not by nominal headcount.

**07**

Control through forecast and action; close through evidence and transfer.

# Assessment Readiness Check

### Knowledge

Can you explain the small-project elements, plan requirements, tools, risks and stakeholder identification?

### Application

Can you produce and interpret the charter, baseline, quality analysis, risk register, RACI, resource plan and control evidence?

### Judgement

Can you choose a method, explain the trade-off and recommend a proportionate action?

### Evidence

Can you trace each recommendation to the case facts and an approved project artifact?

# Support

### Email

enquiry@tertiaryinfotech.com

### Telephone

+65 6100 0613

### Website

www.tertiarycourses.com.sg

### LMS / TMS

https://lms-tms.tertiaryinfotech.com

# Thank You!

Effective Project Management for Small Projects

TGS-2020505545 · Tertiary Infotech Academy Pte Ltd · UEN: 201200696W